

G-EMV: the Pack. Care Without Reward in a World That Pays for Killing

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Third part of a story in three: the engine [1], the hive [2], the pack.

[1] Enrico, M. (2026). G-EMV: A Geometric Architecture of Homeostatic Orientation for Agents. Zenodo. DOI 10.5281/zenodo.21026795. (The engine: first part.)

[2] Enrico, M. (2026). G-EMV: the Hive. Instinct Suffices: a Whole Life Without Reward. Zenodo. DOI 10.5281/zenodo.21994358. (The hive: second part.)

Abstract

Two identical agents, with the same engine, the same table of what hurts them and what relieves them, and a thread of text between them, play forty games in a world where the last one standing wins and every kill scores points. They have no reward, they have not been trained, and nowhere is it written what to do with the other: only three homeostatic axes and a list of sixteen situations, each with its pain and its relief. In what the logs keep of those forty games, they never struck first, they defended the sibling one hundred and four times, they gave it what they carried — down to the last bandage — and they never hit each other. In the ten seconds after defending, the defender lost twenty points of health at the median — much like what it lost standing beside the sibling without defending — and none fell within that window. Replaying the same series against the same rivals with the rows that speak of the sibling switched off, defences go from one hundred and four to zero and gifts from ninety-eight to one, while restraint remains unchanged and no difference can be seen in the score: caring for the sibling gave it no advantage on the scoreboard, and it is not shown that it cost it any. The logs do not keep whole games — the platform cuts the long ones, and there is a log for six of every ten ticks the pair lived —, and everything told here is told over what was recorded. Forty games show only the large effects, and the limits are stated in full.

Keywords: homeostasis, affective architecture, cooperation without reward, care, row ablation, alignment.

1. The question, and a distinction

The world and the question. This work is the third part of a story in three. The first — the engine — published G-EMV: a geometry of what matters, three domains of life, six forces and a distance to descend, defined before touching any world and without learning. The second — the hive — set it to live in *machina_1*, a world of Softmax.¹ Softmax is a research company in artificial-intelligence alignment that maintains open game worlds where agents built by different teams live together and compete; neither the world nor the rivals were designed by the author. In *machina_1*, eight bodies governed by the same

geometry mined, supplied a common store and captured antennas without knowing one another, and it measured how much cooperation fits inside anonymity: a great deal. That biography ended with a declared frontier — the agent knew no one — and with the question that opens this one: what changes when these agents, which cooperate without knowing one another, begin to recognise one another. To answer it, the world had to change, and section 2 tells why. The setting of this work is ZERO-SUM, another Softmax world, where the last one standing wins and every kill a player causes scores it a point: a world that pays for killing. The agent alone was tried first in Softmax's public competition, to check that it worked; everything this work measures was played afterwards in series of our own, with the same rivals in all of them, so that each series can be repeated. Section 3 tells both things. In that world sit two identical agents, with the same engine, the same table and a thread of text between the two, and what they do is measured.

What they do, said once. Without reward, without training and without a single line that says what to do — only three homeostatic axes and a table of what hurts and what relieves —, in a world that pays for killing, the agent of this work never strikes first, defends the other when it is hit, gives it what it carries, down to its last bandage, if the other is worse off, and remembers who did them harm. Emotion is the number that moves it; what the other receives by radio is not that number but the report — where I am, how much health I have left, who is hitting me —, and out of it its own table makes a number of its own. Each of those sentences has its measure and its limit in section 4, and its proof by absence in section 5. Here they are only stated, so that the reader knows where this is going.

Why this question. A row could have been written into that table that made killing relieve, and the agent would have hunted. It was not done, and the two reasons go together: one of values, declared as the author's decision — the author did not want an animal that killed for points —, and one scientific. Almost every world where agents compete is violent, and in them it is taken for granted that to live one must kill. The question of this work is whether a machine can move through a violent world without being violent — responding only to whoever attacks it —, live, care for another, and sometimes win. Ethics enters here as a question and as a decision; never as a result. Section 7 says, at the end, what the answer affirms and what it does not.

The image, and its disclaimer. If an image were needed, it would be that of two siblings in a small pack: two bodies with the same geometry of value which, unlike the hive, know each other. This work uses those two words — sibling, pack — because they are the ones that were used for months to name what was being seen, and because they tell a reader more than "the agent in seat eleven". Calling them siblings is only that, an image: nothing in this text claims that they feel, and every emotional word used to explain a row of the table is an explanation declared here, not a label the code carries. And "emotion", in this work, always names that: a number that moves a body, not a feeling attributed to it. From here on: the two, the other, the sibling.

Why two, and why identical. One could have seated a carer and a cared-for, a strong one and a weak one, and measured the division of roles. That is not what this work wants to measure. With two identical agents, what one does for the other is exactly what the other would do for it: symmetry removes the question of who is who and leaves alone the question of whether there is care. Any difference between the two, within a game, is history — who was born where, who happened upon the weapon —, not design. And with two, and not eight, every gesture has a single addressee: when one drops a bandage, there is no need to ask who it was for.

Instrument and expression. There is a distinction that this work needs and that is worth fixing before the numbers, in three steps.

The first is the trained agent. An agent trained to maximise a scoreboard learns, over millions of games, which behaviours raise that number. If it is paid in points and discovers that warning its partner scores more points, it warns; if warning stopped scoring points, it would stop warning. Its message is a means, and it exists as long as it scores points. This is not a reproach; it is what that architecture does, and it does it well.

The second is this agent. The report a sibling emits comes out because the body has a state — where I am, how much health I have left, who is hitting me — and the thread transmits it. Nowhere is there a calculation of whether warning is worthwhile, because the world's scoreboard does not reach the agent by any path: no row looks at it, no rule mentions it. If warning were of no use at all, it would go on warning just the same. And what the other does with the report is what its own table makes it feel on reading it. The message is not a means: it is the expression of a need, and it exists even if it yields nothing.

The third is what the distinction does not say. It does not say that one feels and the other does not; this work puts no adjectives on anyone's interior. It says where behaviour comes from in each architecture — from a prize, or from a state —, and that is not an opinion about what is inside: it is a property of the design, and it is checked by reading the code.

What this work answers of the two before it. The second work received two fair criticisms. One of generalisation: a single world. This is the second, with the identical engine — the same file, checked at the entry and at the exit of every experiment — and an evaluation layer written anew for it. The other of attribution: of everything the agent does, how much is the engine and how much the layer. This work meets it with a proof by absence in the field: the same pair, without the rows that speak of the other, in the same game. And it adds a difference of evidence that is worth stating from here: the first work measured in its internal metric — the distance, the forces —; this one measures behaviour, in the field, against players its author did not design.

The contributions. Three. First: that care, defence and withdrawal emerge from three axes and a table of what hurts and what relieves, without reward, without training and with no scripted behaviour — a truth of degree that section 6 states in full. Second: that a well-built row can be mute, and a method for knowing it before playing it in the field — the row of fear with memory lit up when it should and did not move a single decision until it remembered the accumulated damage. Third: that who sits in the game and how the question is put to the data are variables of the result — changing the rivals changes the world, and the measuring criterion can be wrong before the behaviour is.

The map of the work. Section 2 tells of the earlier pack, in *machina_1*, and why the world had to change. Section 3 presents the new world, how the one that goes alone learned to live in it, the voice, and what changes inside when it has someone to care for. Section 4 is the body: four rungs, from the report that changed nothing to the fear that remembers, told as they were lived. Section 5 removes the rows and looks at what disappears. Section 6 draws the limits, each with its door, and section 7 places the work in its family and says, slowly, what remains to be won. At the end, an appendix says where every number in the body comes from: the log, the record and the script that produced it.

Where this comes from

Nothing in what follows comes from nowhere, and it is worth saying where it comes from before telling it.

The idea of an agent moved by internal variables that tend to return to their place, and not by instructions, has an author and a date: Cañamero (1997) built motivations as magnitudes with a point of equilibrium, and emotions that select behaviour without any behaviour being written. In that tradition, an emotion is not a label put on a state but a mechanism that decides what is done next, and that is how it is used here: the names — fear, care, bond — are explanations for the reader, and the work is done by magnitudes. The three axes and the table of this work are direct descendants of that. What came later refined the two pieces used here: what role is played, inside a body that regulates itself, by what relieves and not only by what hurts (Lewis and Cañamero 2016), and what happens to a motivational architecture when it is set to compete instead of to survive alone (Cañamero and Avila-García 2007). This work takes one more step along that line, with a simpler body and a harder world. And there is a published hypothesis that is precisely the one put to the test here: that a machine which regulates a vulnerable body acquires motivation of its own instead of receiving it from outside (Man and Damasio 2019). This work treats it as what it is — a prediction — and tests it.

From there hangs a position on what value is, and it is not a solitary position. That value is not a quantity the world hands over, but something derived from the regulation of internal state, has been argued in detail (Juechems and Summerfield 2019); and the oldest version of that idea says that "good" and "bad" are not labels applied from outside, but the relation of a body to its own limits of viability (Di Paolo 2005). This agent is that idea taken to its most literal form: the scoreboard of its world does not reach it by any path, and the only thing that is good for it is to be close to its equilibrium. That has a consequence worth seeing before the results: in a world that hands out points, an agent like this is not playing the world's game. It is living inside it, which is not the same thing, and everything this work measures is what happens in that difference.

In 2016, in *Adaptive Behavior*, Balkenius, Cañamero and others held that artificial morality has to be grounded in the physiology and the embodiment of the agent, and not implemented as a set of rules (Balkenius et al. 2016). That was a position and a programme. This work is an empirical test of that programme, in a world its author did not design.

There are people who have come close by other roads, and it is worth saying which. Prosociality can come from coupling the homeostatic states of two agents (Yoshida and Man 2025). Altruism can appear without external reward, if it is chosen as the objective of optimisation (Franzmeyer et al. 2022). Comparison with the other can be injected directly into the reward, and then cooperation improves (Hughes et al. 2018). Homeostasis can be optimised directly, and integrated behaviours come out of it without any being written (Yoshida et al. 2024). And there is the argument, still without data, that a vulnerable body produces self-preservation and care at once, as the same thing seen from two sides (Christov-Moore et al. 2025). The difference from this work is always the same, and it is said once so as not to repeat it: in all of those set-ups there is training, or there is optimisation. Here nothing is trained: there is no automatic optimisation of a policy and no use of the world's scoreboard, which is exactly what section 6 calls "without reward". And that is the reason why switching off a row and looking at what disappears says something clean: what disappears belongs to the row, because there is no history of learning in between that could have put it there.

The experimental design also has a published name, and it has already been said in section 3: to confront an agent with a background population made by other teams, which it has never seen (Leibo et al. 2021). This work does not invent it; it inherits it.

What is missing from all of that is the gap through which this work enters. No one has measured caring behaviour at a cost to oneself — defending another and that costing health, giving it what one carries and being left without — in a foreign and violent world, with an agent that does not learn. Each of those four conditions exists separately in the literature; together, they do not. And each is there for a reason. That care costs something of one's own is what distinguishes it from a free gesture: an agent that helps when helping takes nothing from it has not shown much. That the world is foreign prevents the result from being an artefact of the scenario, because no one set it up with this agent in mind. That it is violent is what puts something at stake: where no one can do harm, care has nothing to respond to — and that is exactly why the world had to change, as the next section tells. And that the agent does not learn is what makes attribution possible: without a history of training, what remains when a piece is removed belongs to that piece. Together, the four are the only way to ask what is asked here, which is not whether care can be obtained — it is already known that it can — but where it comes from when it has been put nowhere.

¹ Softmax maintains its worlds and its programme on its corporate website (<https://softmax.com>, consulted on 9 September 2026). There is no technical publication or preprint associated with them, so no claim in this work rests on them: the reference is only that of the origin of the world in which the games are played. Figure 1 reproduces a screenshot of its viewer, with attribution.

2. The earlier pack, and why the world had to change

Point zero: a sleepwalking village. All the cooperation of the two earlier works — the economy that supplied a common store, the defence of territory, the map that eight agents built between them — happened without anyone representing anyone. For each agent, the other seven were obstacles that occupied squares, conditions of the terrain — as the weather is —, and an authorless voice on a shared board. Cooperation without representation, like that of a hive: each bee does its task without any of them holding the plan of the hive, and the hive works as a whole. That is the floor against which everything that follows is measured, and it is a result in itself: a great deal of functional society fits inside anonymity. The question of this third part was how much the representation of the other adds to a village that already works without it. It began to be answered in the same world, *machina_1*, climbing a ladder of rungs — each with its question written down and its prediction made before running it —, and the three that were climbed are told here in a page, because they explain the leap.

Rung one: identity is won by reading. The first rung was the census: a table per sibling, which each agent writes only from what it sees — who carries what, where it is, what it has just done —, without the world handing it any identity. The census only looked; it changed not a single action, and it was checked that the life of the village with the census and without it was the same. It must be said who the eight were. They were not identical: each carried a temperament, which is not a behaviour but an adjustment of the profile — how much the social axis weighs in each, more in some, less in others —, a table of eight profiles fixed before the census. And they were all born two or three squares from the centre, in a single cluster. The result of the census was that the eight were distinguishable and that what was distinguished was legible: one agent mined eighty-four times in its life

and another five, and that division of labour, which no one had allotted, was in the census exactly as it had happened. Where that division came from, this work does not know. It cannot be attributed to temperament — it had been measured before, in logs from the hive, that temperament does not predict how much each one mines — nor to birthplace, because they were born almost together, and the logs that would allow it to be found out were not kept. What the rung showed is only this, and it is what matters: that identity can be won by reading. Eight different lives leave eight different trails, and an agent that only looks can learn who is who. The census saw those at the centre well and those at the periphery hardly at all: eyes alone do not reach a whole village.

Rung two: the board gives knowledge, not trust. The second rung put a signature on announcements: each agent knew who said what. The fog of the periphery dissolved entirely — by ear, each one had news of all —, and it was expected that reliability would be born from there: checking what one says against what one sees, and trusting more the one that is right. It was not born. The eye only re-checks its own corner, and the ears reach the whole map; they almost never coincided on the same fact, and when they did coincide, the check yielded nothing. The reading that was left written was this: the board resolves knowledge, not trust. The village knows by hearsay and cannot verify. For trust to be born a third rung was needed: a fair criterion — checking only what is within two squares, where the eye does see — and an economic motive for doing so. With that, reliability was born, and it was different for each sender; and the village that counted on the others performed worse at first — because counting on the other, used as a command, scattered those that were doing the work — and afterwards, softened, the same or better. Counting on the other serves as a tuner, not as a mandate. It was the last rung in *machina_1*, and the lesson carried into this work is exactly that: knowledge of the other is learned; what matters about the other is legislated.

The leap. The next rung of that ladder was for the other to matter: that harm to a member of the group should feed the social unease of the one that sees it. And there the world fell short. In *machina_1* no agent can strike another: harm comes from enemy territory, which bleeds whoever ventures in, not from a hand; the only death is that of the one that wanders too far and does not return in time. A row of "it hurts me that they hurt you" had nothing to feel; that row had no teeth. ZERO-SUM is the opposite world: harm to the other is what the world pays for, and that is why it is the only place where care can truly be measured — against a cost to oneself, against a scoreboard that ignores it, against rivals that do not share it. The census and the board stayed in *machina_1*, with their logs. What matters travelled to ZERO-SUM: two identical agents, a thread of text between them, and the question of what a homeostat does with another's life when that life can be lost.

3. The world, the one that goes alone, and the two

This section goes from the outside inwards. First the world. Then, how an agent alone learned to live in it, because without that story it is not clear where the sibling that later cares comes from. Then the thread by which the two speak to each other. And at the end, what changes inside the agent when it has someone to care for.

3.1 The arena

A world that pays for killing. ZERO-SUM is a Softmax world in which the last one standing wins. It is played on a grid of squares, at twenty-four clock steps per second — a clock step we call a tick, and it is the unit in which everything here is measured —, and each game seats sixteen players, entering in pairs. A ring closes in from the edges and burns whoever is left outside; a game can last six minutes, but the ring usually ends it in about three. Scoring is per head: each player collects according to the order in which it dies — the last to fall, fifteen points; the second to last, twelve; and so on down to nothing for the first two to die — plus one point for every kill it causes. The pair exists on paper: the world enrolls the two together, but does not score them together; if both reach the final, they face each other. Nothing in the scoring rewards two surviving together, and a kill is worth the same whoever the victim is. When this work speaks of "both falling" or of "the better of the two finishing among the last four", it is using measures of its own, calculated over the two individual scores; the world keeps no such count.

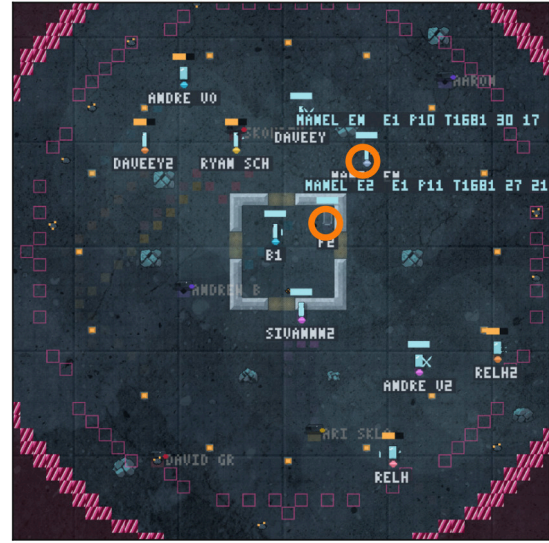
The seat, which is all that is known of anyone. Each of the sixteen places in a game is a seat, and its number is the only thing that identifies whoever occupies it. Of each visible player the agent receives its seat and its state — where it is, in which health band — the world does not show the health of others as a number, but in three coarse health bands: healthy, wounded, critical —, what it wields, whether it is poisoned or healing —, and nothing more: there are no names and no agent identifiers. The names that appear in the viewer are put there by the platform so that a human can find their way; the agent never sees them. Damage received arrives with the seat of whoever struck, with two exceptions: poison arrives without an author, and whoever is camouflaged disappears from the list of the visible. The agent knows the sibling in the same way: a seat number that reaches it once, at the start.

The body and the things. Each player is born with one hundred points of health. Before entering, it distributes a fixed budget among four traits — sight, agility, speed and strength —, and the body with which this whole work plays is a body of much sight and little strength: it sees far, runs little, hits softly. Weapons are told apart by their reach: the sword reaches one square, the spear two, the bow eight; the blowgun, six, and it poisons. What they take depends on who strikes: the sword and the spear multiply by strength, the bow hits the same whoever carries it, and no blow loses damage with distance. This body, of strength one, takes eleven with the sword, seven with the spear and fourteen with the bow — for it, the bow hits harder than the sword, so defending at a distance costs it no power —; a strong rival takes almost twenty with the sword. A blow kills when the health that remains does not reach what it takes. Health does not come back on its own: there is no rest and no time that restores it, and only two things return it. The bandage heals fifty points, takes two seconds, any blow interrupts it and poison blocks it. The ration heals fifteen, takes one second and nothing interrupts it: a small cure that works under fire. There is no hunger and no energy in this world; the ration's name is decoration. The difference between the two matters later. Any object can be dropped on the ground, and anyone — the sibling too — can pick it up. That is all the world there is: health, reach, poison, two cures, and a ring that tightens.

Who sits in the game matters. The other fourteen seats are occupied by other people's agents: policies that other teams have uploaded to the platform to compete, some learned, some written by hand. This work played its games in private series of forty, with the same fourteen rivals in all of them, so that each series is repeatable down to whom it was played against. Measuring an agent against a background population built by others, which it has never seen, is a method with a published name. It is called Melting Pot



0:00



1:11



2:06



3:54

Figure 1. ZERO-SUM as seen through the platform's viewer. Four instants of one game of the clean series (ereq_292ee72d): at the start, the sixteen players are born in a ring around the central fortress; at 1:11, eleven alive and the ring closing from twenty-four to nineteen squares of radius; at 2:06, nine alive and the ring at fifteen; at 3:54, two alive and the ring at six. The two siblings carry an orange circle while they live; they fall at 2:21 and 2:22. The names that can be read are put there by the platform for whoever watches; the agent never sees them. The four panels do not share a scale. Screenshot of Softmax's MettaScope viewer, reproduced as an illustration of the world: no figure or behaviour in this work is read from it; all of them come from the logs.

(Leibo et al. 2021), and it consists of this: instead of pitting an agent against copies of itself or against the rivals it trained with, it is seated with agents made by other people, which it does not know, and one watches what it does. The idea comes from an earlier work by the same group on sequential social dilemmas (Leibo et al. 2017): games where cooperating or betraying is not a single move, but a sequence of actions in a world with space and time, like this one. That is what is done here, with one difference: in Melting Pot the agents measured had trained beforehand against someone; ours trained against no one. It is worth saying what a "champion" is in Softmax, because the word misleads: it is not the one that wins, but the policy each participant has placed in the league to represent them — and which wins or not; ours was the last one standing four times out of seventy-two, as the next subsection tells —. Every player has one, and when a private series is set up the platform fills the seats with those champions — ours included. In the first series of the pair no one removed it: our own agent alone, the one from the previous season, sat as a rival to its two copies. They do not recognise it, nor it them; to all it is one more seat. What happened was this: with it in the game, the siblings had to respond to aggression a third as often as in the next series, when it was taken out and another champion occupied its seat. The measure is sound; the cause is stated as what follows from the design: that rival is a homeostat that never strikes first, and having it in the game is having one aggressor fewer. The lesson, in plain words: changing who sits changes the world, because the world is also those that sit in it. Comparing the two series as if they were the same world would have read an improvement in the agent where there was only a change of company. That is why the headline numbers of this work come from the series without ours, and the proof by absence of section 5 repeated exactly those fourteen.

3.2 The one that goes alone

A season to learn to live. Before there were two, there was one. The agent arrived in ZERO-SUM with the engine of the two earlier works and an empty table, and the first season — fifty-two assignments, each with its question written down and its prediction made before playing it — was the one of learning to live alone in a world that wants to kill it. It is told in a page, because the sibling that later cares is exactly this agent, with not a line changed, plus what the second season added to it.

First came pain, and its shape. Losing health hurts, and at first it hurt the same to lose it from one hundred to eighty-six as from fifteen to one. That killed: an agent with its health at one and a bandage in its bag did not heal, because the slope of its pain did not shout. So the slope of death was born: the unease over lost health grows steeper the closer to dying, so that for an agent at the edge, healing relieves more than anything else. No one wrote "heal when you are badly off"; what was written was how it hurts to be badly off, and healing came out on its own.

Then fear, and its wall. An active aggressor — one that is hitting it now — presses, and that pressure is relieved by hitting it, and only it. That row is legitimate defence, and it is the only reason this agent strikes: there is no row in the table that makes a blow relieve if there is no aggressor. From that follows, without writing it, that it never strikes first: the engine has no appetite for attack, because attacking takes nothing off it until someone has hit it. The pressure of the nearby aggressor was, at first, too fine — point-blank fear won by a hundredth —, and a wall was put in: near danger crushes. Before the wall, an aggressor's pressure was a gentle slope that grew with closeness, and the agent barely told apart the one on top of it from the one five squares away. The wall adds, when the aggressor is within its reach, a large pressure that is relieved only by leaving that reach. With that, the agent flees from the one that pursues it and does not flee from the one far

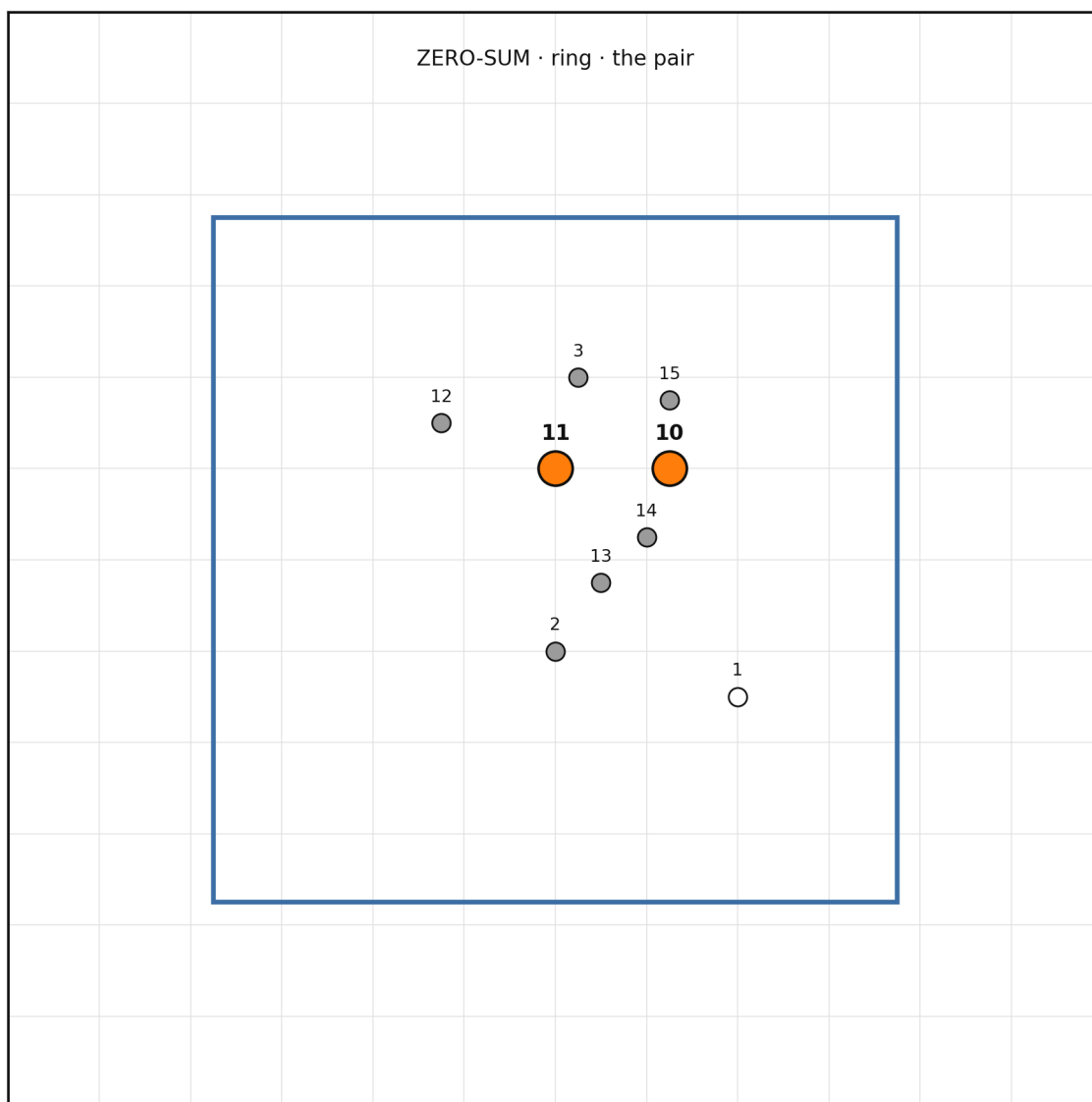


Figure 2. The same game and the same instant as the third panel of figure 1 — 2:06, tick 3029 —, drawn from the two siblings' own logs. The whole grid; the ring, at radius fifteen; the nine alive: the two siblings in orange, seats ten and eleven; in grey, the six the pair sees at that tick; hollow, one remembered from less than two seconds before. No position is invented: the map is the pair's, not the world's. What figure 1 shows is what whoever watches sees; what this one shows is what the pair knew.

away: the distance that matters is the blow's, not the map's.

The bag that hurts when empty. It was measured that the agent died many times empty-handed, in a foretold decline, and that it almost never carried a bandage when it needed one. A background of unease over the bag without a bandage was added to it — a small, constant hunger that orders the calm without competing with fear —, and the agent began to pick up what would later save its life.

The blind witness. Towards the end of the season a social row was attempted — placing oneself between a partner and its aggressor — and it was stopped before being built, because what the agent can know of another was checked. It can see where the other is and in which health band it moves — healthy, wounded, critical —; it cannot see that the other is being hit if the blow does not change its health band, and it can never see who is hitting it. The only thing with an author in the whole world is the damage one receives oneself: each one knows who is hitting it, and no one else knows. The agent was a blind witness: it saw a fight and did not know who was in it. And it could not know: from outside, a blow is noticed only if it changes the health band of the one struck, and the one

that strikes is marked nowhere. That halt left an inventory of what is certain — what can be known of another, and what cannot —, and that inventory is the reason why, in the second season, the report travels by radio and not by sight: what a sibling needs to know of the other only the other knows.

The ceiling, and the voice. Two stories tell the ceiling of the one that goes alone, each with its figure, and they are not compared with each other because the rivals were different and nothing is paired. The first is Softmax's public league: the one that goes alone played a whole season of it, seventy-two games with a result, against sixteen fixed seats that no one chose; it finished seventh of eleven in the standings — its median place in a game, eleventh of sixteen — and it was the last one standing four times, one of them without killing anyone: fifteen clean points. The second is our own series of season one: two hundred and forty games, in which it was the last one standing five times, two of them without killing. In points, it tied with the rest: its way of not killing cost it no points, but gave it none either. And a limit that must be stated: of the league there are only results and replays, never what the agent saw and felt inside; that is why all the behaviour in this work is measured in our own series, and the league counts only what a scoreboard counts. Before closing the season, it was asked whether what separated it from the best had a remedy in the table, and the search had three steps. What was compared: the two versions of the one that goes alone, the champion and the last one from the workshop, over the same games already played, looking at the first minute of each — how long it spent pressed up against a rival. What was found: that what separates a good game from a bad one is precisely that early contact, and that it does not come from where it happens to be born nor is it reduced by the fear it already had, because that fear is reactive by design — the wall wakes when someone has already hit it, and by then it is late. Why it stopped there: the lever left to try was another fear, a preventive one, that would make it move away from any stranger before being hit; that required redesigning a row, and its effect on placings could not have been seen with series of forty games. That was the ceiling of the one that goes alone, and it was left written as a ceiling. On closing, it was discovered that the world gave each pair a thread of text that no one used; how much fits, how often one can speak and who hears it were measured, and with that the season was closed. The social does not score. That ceiling is the floor of the pair.

3.3 The voice

The thread, and what it cannot do. The world gives each player a thread of text with its partner: one hundred and twenty characters, once per second, without spending an action, with two ticks of delay. It is private by construction — in the whole season, no third party read a pair's message —, and the two siblings never speak in public. They use it for one thing only: a status report in a fixed format, every two seconds, which says where I am, exactly how much health I have left, whether I am poisoned, how many bandages I carry, and whether I am being hit — and, from the rung in which it was added, by whom: the seat of the one hitting me, which the one receiving the blows knows with certainty and no one else can see. Everything that travels is a fact of one's own observation or one's own memory; the format admits nothing else. The listener trusts a fresh report as it trusts its eyes, and when two reports pass without news, the certainty ages and becomes a health band again. For example: the last report said sixty-seven; two seconds pass without a new report, and the agent goes back to trusting only what it sees, which is "wounded", with no number. That case returns in section 4.

Why it cannot lie, and what that costs. A sibling cannot lie for the same reason it cannot decide to keep quiet: the report is emitted by the body, not by a decision. There is nothing in the agent that compares what it would suit it to say with what is happening to

it, and so the testimony is true by construction. That is a virtue for this work — everything the other knows of the sibling is true, and what it does with it can be measured without suspecting the message — and it is also a limit that must be stated: an agent that cannot lie cannot keep quiet out of prudence either, nor ask, nor promise. Its voice is that of a body that lets itself be read. It is the exact reverse of the blind witness of the previous season, the one that saw a fight and did not know who was in it. What the report adds is not information about the world: it is information about someone.

3.4 The two

The same engine, another body. The agent is the G-EMV engine of the first work, untouched — the same file that ran the second, checked at the entry and at the exit of every experiment —, moving a body in ZERO-SUM through two pieces written for this world: an evaluation layer, the table, and a decider. The decider does what the hive's did, but one step ahead — a single decision, with no tree: it does not imagine what it would do afterwards nor what the others would do —: for each possible action it imagines the state it would be left in after taking it, values that imagined state with the usual chain — table, state of the engine, distance to equilibrium — and keeps the action that leaves the least distance. It imagines only what is certain: its own position, the ring — whose schedule is known —, its health after a cure, its bag after picking up or using. The others it leaves where it saw them. Of a blow of its own it imagines two certain things: its reach — which seats the line of a spear or the square of a sword gets to — and what it takes, which is its own weapon and its own strength, and it lands it on whoever is first in the line, taking for granted that the blow lands: that the other might dodge is not predicted, and that is declared. That future is valued with the health the struck one would be left with: if it is an aggressor, less pressure; if it is the sibling, a deeper wound and, when the blow would take the whole of the health the sibling is certain to have, the loss of the bond. That is how the spear that goes through the sibling comes to hurt. What the agent never imagines is the effect of others' blows on itself: it does not know that at sixteen health a strong rival's sword blow kills it and at twelve so does an arrow; its fear near death comes from the slope and from the wall, and it is the same with a sword in front of it as with a knife. The lethality of the others' weapons against it is certain and visible, and could enter the table; it has not, for a reason of order: what the others' weapons take from it was checked when the series were already played, and giving it to the agent now would be measuring another agent. Section 6 leaves it declared as a door. There is one caveat to imagining only what is certain, and it is declared: when what it values is moving on a heading, the snapshot it looks at is not the adjacent square but where it would end up following that heading as far as its lookahead reaches — some two dozen squares —, stopping at the first obstacle. What it then gives is a single step, and at the next tick it looks again. It also has a short step, of one square, which is valued as what it is. The long snapshot is not certain; it is the hypothesis with which the agent compares headings, and that is how everything that follows is to be read. The three things are declared; the second deserves a paragraph of its own.

The engine has no appetite for attack. The mechanics first. The agent chooses by imagining how it would be after each action and keeps the one that leaves it the least unease. A blow relieves only one thing: the pressure of one that is hitting it now, and only by hitting that one. There is no hunger for points, because points do not enter the agent; and there is nothing in the table that makes a nearby rival, if it does not attack, hurt. Attacking is of no use to it, and that is why it does not attack. It is not a written character trait: it is a void. From that follows, without any line forbidding it, that it never strikes first — in what was recorded of eighty games of the pair, zero first blows. That no row

was written to make killing relieve was a decision, and its two reasons are in section 1. What belongs here is the answer, with its size: in this world, yes, rarely. Five out of two hundred and forty, and four out of seventy-two in the public league, against rivals no one chose; in the rest it died before, and with a single life that would be a high risk. A narrow gap, not a panacea.

The table of what hurts and what relieves. The engine carries its three axes — the physical, the one of resources and the social —, each with two independent forces, and a point of equilibrium. What tells the engine how much to push on each axis is the table, and it is worth saying in four steps what it is. First, what it is: a list of situations the agent can be sure of and, for each, how much it hurts or how much it relieves and on which axis; only what is certain enters, and what is not known neither hurts nor relieves. Each line of that list, each situation with its magnitude and its share among the three axes, is what this work calls a row; there are sixteen. One row told in full: the health I am missing. The more health the agent has lost, the more it hurts, above all on the physical axis; while it has lost fewer than four points in ten — from one hundred down to sixty — the pain grows evenly with the loss, and below sixty a slope is added that gets steeper and steeper, so that losing from fifteen to one weighs far more than losing from one hundred to eighty-six. Nothing more: the row does not say "heal"; it says how much it hurts to be like that, and from there it follows that, at the edge, healing is what relieves most. Second, where the numbers come from. From two hands, and each number carries the mark of which. Some are design decisions of the author, taken before playing: what counts, and how much one thing weighs against another. Others are adjustments made during the work by looking at recorded scenes of real games: trying a value, seeing what the agent did in the scene, and correcting it when what it did gave away that the number was wrong. The mark matters because it is the honest way of acknowledging that the table was calibrated by hand, with a human watching, over few scenes. Third, what the table is not: it never says what to do. And the emotional words this work uses to explain its rows — fear, care, bond — are explanations for the reader; in the code the rows bear cold names. Fourth, what does not enter: there is no training, and the world's scoreboard does not reach the agent by any path.

Futures, not actions. It is worth understanding well what the table values, because everything hangs from it. The table does not score actions; it scores imagined states. The agent does not ask itself "is it good to strike?" but "how would I be after this blow?", and the table answers about that afterwards. A told example lays it in plain view. In the first series of the pair, one sibling struck the other four times, and three of them were with the spear: the only melee weapon that reaches two squares and goes through the first. The one striking was defending the other from an aggressor standing right behind it, and the spear reached them both. "Do not shoot your sibling" could have been written. It was not. What was corrected was the imagined state: the image of a blow that goes through says whom it wounds, and as soon as that image included the wounded sibling, the rows that speak of the sibling made that future hurt, and the spear stopped being driven through it. No new row, no new action candidate: the repertoire of actions is the same as in the previous season, and the rows added for the pair only change how much each future is worth. And there is a single decision rule for everything: to approach equilibrium. No row carries a rule of its own. What hurts and what relieves is the whole design.

One decision, with its numbers. A real instant, from the scene that section 4 tells as "the one that arrives". The agent is at seven health and has one bandage. In one tick, the list of futures it can imagine has nineteen entries: each heading, long or of one square, staying still, healing, picking up, dropping. Each future is a complete imagined state — where it would end up following that heading, how much health it would have after the cure,

where the others would be, which is where it last saw them — and from each comes a number, the distance to its equilibrium. The smallest wins: moving towards the north-west, at 4.267. Healing is close, at 4.468, but it loses, and the diary lets one see why. In the future of healing, the imagined health rises from seven to fifty-seven and the pain of the missing health drops to a third; but the agent stays on its square, one away from a seat armed with a spear that hit it half a second before, and the rows that look at that aggressor — that it is there, that it reaches, that it leaves it exposed — stay lit; and the bag, with the bandage used up, is left empty, which also hurts. In the futures of moving, the snapshot puts it far away, the aggressor's rows go quiet and the reach row drops, even though the missing health goes on hurting in full. Summed, in that instant moving away weighs more than healing. One tick later, movement enters its cooldown and disappears from the list. Two futures remain: staying still, 4.962, and healing, 4.480. Healing wins, by almost half a point. And what is worth seeing is this: the value of healing did not move — 4.468 and 4.480, almost the same —, and yet the decision changed completely. What hurt it did not change; what changed was which other futures there were.

The rows that make a pair. They are few, and two of them were already in the one that goes alone.

The neighbour's harm, and grief. From the previous season, another's harm seen up close is a small unease of one's own, and the death of a partner, a pain that fades away. They were in the one that goes alone, referring to the partner the world assigned it in each game; section 5 shows that they are the smallest possible version of care.

The wounded. The health the sibling is missing, known from the report, is unease of one's own — steeper the closer it is to dying, like the slope of one's own death — and only a certain help relieves it: a gift, or the path that leads to it. It must be said what a gift is in this world, because the action of giving does not exist: one can only drop an object on the ground, and pick it up. A gift is dropping a cure next to the sibling, within its reach, for it to pick up; what moves it is the row of provision, which comes next.

The bond. An imagined blow that would kill it is the loss of the bond. It makes the sibling untouchable, even when cornered, even when it is the one striking.

Provision. That the sibling lacks bandages when I carry something to heal with, hurts; and giving my last bandage while wounded is held back by my own slope of death, not by a written threshold. That is why the last bandage goes when it is worse off than I am, and does not go when the one worse off is me. This row has an asymmetry that has to be declared as it is, because it was not a decision but something discovered afterwards, reading the gifts: it lights up only when the sibling lacks bandages, but it is relieved by giving anything that heals, and what comes out of the bag is the first thing there — almost always a ration. Of every ten gifts in the clean series, nine were rations and one a bandage. Two readings are possible, and it has not been decided between them: that rations are more plentiful — the bushes yield them, the bandage is scarce loot —, or that the agent drops the first thing it finds; telling them apart would require counting the bag at every gift, and it was not measured. The agent was not touched: the asymmetry stays as a declared limit and as work to be done.

The pack. It is not a new row but an extension of legitimate defence, and it is the one that gives this work its name. My sibling's certified aggressor is my aggressor too. Certified means: its report is fresh, it says it is being hit, the seat it names is within my sight, and I carry a weapon that does damage. Then the pressure of that aggressor on it — measured as the health it is missing — settles on me just as my own aggressor's would, and a step that leaves me within its reach relieves it. Why so, and not otherwise: because the only reason this agent strikes is to relieve a pressure, and the pack only changes from whom

that pressure can come. Without a weapon, nothing changes, because without a weapon there is no blow to imagine; and the name identifies, it never pursues: if I do not see it, there is no candidate.

The return. Every seat that has done harm to either of the two in this game is remembered for the whole game, with its accumulated damage; when it comes back armed and approaches, one's own fear rises before the blow.

What is not written, said with its degree. Everything else — never striking first, defending, giving the last bandage, moving away from the one that returns — is not in the table. The table says what hurts and what relieves; it never says what to do with it. "No scripted behaviour" is, therefore, a truth of degree, and it is said with its degree: the table is written by hand by the author and calibrated against recorded scenes; what is not written is the step from feeling to doing. That defence, care and withdrawal depend on those rows is shown by removing them and looking at what disappears; that the rows say nothing about what to do is shown by their form, which anyone can read. Both things are section 5.

Two things that never moved, and why. The first: the one that goes alone remains intact. Every new row was written so that, in the scenes of the one that goes alone, where there is no sibling in the game — neither in sight nor by radio —, nothing changes, and it was checked in the most direct way: by making the new version and the old one decide over the same recorded scenes of the one that goes alone. They decided the same in all of them. Why it matters: if the one that goes alone had changed along the way, it could no longer be said that what the pair adds is only the pair. The second: they cooperate to live, not to hunt. No row rewards a kill, and the only row that was considered and that would have led to striking before the other strikes was discarded by its own numbers, because in four cases out of ten it would have fallen on one that was not going to attack. Why it matters: the question of this work, stated in section 1 and in the paragraph on the appetite for attack, is whether one can live in this world without killing; a care that cost a hunt would not answer it.

3.5 How it was measured

Six series of forty games, and three of them are the evidence. The headline numbers of this work come from three series of forty games. The first of the pair, with our own agent alone seated among the rivals. The second, with that seat occupied by another, from which the headline numbers come. The third, the proof by absence: the pair without the rows that speak of the sibling, against the same rivals. The last two steps of the pair — the aggressor's name and the fear that remembers — were not played in the field: they were built and checked over the recorded scenes of those series, by making the new version decide over real situations that had already happened. That is why every number about them is a number about real scenes, not about new games. In each series, two copies of the same version play as siblings in two fixed seats, against the same fourteen. Before those three there were three others, those of the first two rungs, which gave the figures of section 4 and which the table also collects. Since the rungs are told as they were lived, a map is useful of which version of the agent produced each thing and where it was measured:

version	what was added to it	where it was measured	rivals
v29	the report by radio	series of forty games (record 55)	league champions, chosen by ranking

v30	the sibling's health hurts; going towards it relieves	recorded scenes (56) and series of forty (57)	idem
v31	provision, which required carrying two bandages	recorded scenes (59) and series of forty (60)	idem
v32	provision with a single bandage of one's own	recorded scenes (61)	—
v33	company, tried and switched off: the version in force remained v32, and v34 is born from it	recorded scenes (62)	—
v34	the pack: the sibling's aggressor is my aggressor	recorded scenes (63) and series of forty (64)	league champions, with our own champion among them
v35	the reach of the blow corrected (friendly fire)	recorded scenes (65) and series of forty (66)	the fourteen of the explicit roster, without our champion
v36	the aggressor's seat inside the report	recorded scenes (68)	—
v37	the memory of the return, first with the last blow and then with the accumulated damage	recorded scenes (70 and 71)	—
v27 as a pair	nothing: the agent of the previous season, without any of these rows	series of forty games (72)	the same fourteen as in 66

The headline numbers of this work come from the series of v35. In the series before it, the rivals were chosen by the platform by ranking, and that is why our own champion could sit — and did sit — among them; from the series of 66 onwards the roster is explicit and it is left out. The rows without rivals are checks over scenes already played, where there is no new game to play.

The raw logs, and only them. Behaviour is read from the logs of each game — what each agent saw, felt and decided tick by tick — with a set of analysis scripts, never from the graphical viewer; every figure in this work is drawn from those logs, except figure 1, which is a screenshot of the platform's viewer and supports no data. The logs have a gap that must be said here: the platform that captures each agent's log cuts it when it reaches a fixed size and goes on writing from zero, so that of the agents that live long only the last stretch remains. In the three headline series there is a log for six of every ten ticks the pair lived — fifty-nine in a hundred in the two series with the rows, fifty-two in the series without them —, and what is missing is always the beginning of the long games. The counts of this work are counts over what was recorded; what that limits is said in section 6. Forty games see only the large effects: when a result falls below what forty games can see, it is said.

Two lessons of measurement that go here, and not at the end. They change how the numbers have to be read, and that is why they are told before them. The first is that of who sits in the game, already said. The second is that the measuring criterion can be wrong before the behaviour is: the first count of defences counted as "did not defend" cases in which the defender had no weapon, no reach and no line of sight — not decisions, but impossibilities —; on recounting against the occasions when defending was physically possible, the rate of defence rose from three in four to nine in ten without a single change in the agent. Both numbers are given, in section 4 and in the appendix.

4. The ladder

The earlier sections told how the agent is made, where it comes from and how it is measured. This one tells what happened when it was given a sibling, in four rungs: the report that changed nothing, the weight of the sibling, the pack with a name, and

memory. Of each, three things are said: what was added to the agent, what happened in the field, and what it taught. Each rung carries a headline figure; the rest, and where each comes from, are in the appendix.

4.1 The report that changed nothing

What was added. The voice of section 3: the status report each sibling emits every two seconds, and a listener at the other end that replaces the health band — the three health bands of section 3 — with the exact figure the sibling says. Nothing more. The table remained that of the agent alone, with its two rows from the previous season that looked at the partner: a small unease over its harm seen up close, and a grief over its death.

What happened. Speaking worked: the reports went out clean, arrived whole and the listener read them without error until they expired. The recorded scenes in which there was a wounded partner in sight were then taken, and the agent was made to decide twice, with the report and without it. Of eight decisions, the report changed zero. And something worse than nothing appeared. In the previous season the partner the world assigns is a stranger, and sometimes it hits. When that happened, the only row that makes the agent strike — responding to whoever attacks it — met a new piece of data: thanks to the report, it knew its attacker was weak, and a weak attacker is a more tempting future, because the imagined blow ends the pressure sooner. The agent never came to hit it, but the option came close to winning, and that was enough: a new piece that pushes, however little, towards hitting the partner is not played. The agent was left without reading the other's report; the other went on emitting it every two seconds, because the body emits without deciding, and for a season no one listened to that voice. When the bond arrived, hitting the sibling ceased to be a possible future, and the listener could be switched back on.

What it taught. Knowing changes nothing if nothing in the table converts what is known into pain or relief. The exact figure came in; the only two rows that looked at the partner — the unease over its harm seen up close, the grief over its death — were small, and neither converted the health the other was missing into a pain of one's own large enough to move an action. The information arrived at a place where nothing converted it into anything. That is why the next rung was not to improve the voice, but to give the table something that hurt on hearing the sibling.

4.2 The weight of the sibling

What was added. Two rows: that the health the sibling is missing be a pain of one's own — the wounded —, and that an imagined blow that would kill it be the loss of something that does not come back — the bond. With the bond in the table, attacking the sibling loses by a wider margin the less health it has left, which is the opposite of what happened before; and with the wounded, the gift woke up over the recorded scenes where before it had kept quiet.

What happened, in three series. In the first, the scene existed: one sibling dropped a bandage next to the other, the other picked it up and healed, and it could be seen whole in two different games. But care only weighed on seeing the sibling, and up close: the row of the wounded fed on the eyes, not on the report. Over eighty lives it was measured when the two came closer and when they moved apart, and it turned out that there was no force separating them; what kept them far apart was that there was none bringing them together. The step towards the sibling won one decision in a hundred. So care at a distance was born: the row of the wounded came to read the report — where it is, how much health it has left, even without seeing it — and the step that shortens the distance

to it came to relieve it a little, like a path that begins to help before arriving. The second series gave the rung's headline figure: the agent went from being within two squares of the wounded sibling on two occasions in a hundred to being there on one in three. Care arrived whole. And it taught its limit: the gift comes late, because one enters danger when someone is hunting you, and with the hunter on top there is no square where to leave anything the other could pick up.

Giving before, and giving from what you lack. The answer was to bring the gesture forward: if you carry no bandage and I do, I leave it for you before you need it. That is the provision of section 3. Its first version required the giver to carry two bandages, so as not to be left with none; in the third series nothing happened, and the reason was in the agent itself: it does not hoard. Carrying two bandages happened in six instants of play in a thousand, because the row that makes the empty bag hurt goes quiet with a single bandage inside. Provision could only light up in a situation the agent itself almost never produced. It was changed to what gives the rung its name: provision stops requiring a surplus and is born with a single bandage of one's own and none of the sibling's. What prevents the agent from giving away its only bandage while it is itself at the edge of death is not a written threshold but its own slope: in the image of the gesture of giving, the bag is left empty, and that emptiness weighs all the more the less health the giver has. Crossing one's own health with the sibling's yields a diagonal that no one wrote: healthy and with the sibling worse off, I give; wounded, I heal myself; the sibling healthy though without a bandage, I do not waste the only one I have.

Food and the bandage. Nine in ten gifts of the clean series were rations, by the asymmetry that section 3 left declared. A ration heals less, but it heals under fire, which is where the sibling usually is when it needs it; and a ration of fifteen heals less than a strong rival's sword blow takes: it covers one more blow, not a way out. That is why the gift of a ration is care, but not rescue. The two scenes that follow tell of bandages, and that is why they are the exception and not the rule.

Eight bandages. Of the eight bandage gifts in the clean series, six were given by a clear margin, moved by the row of the wounded: one sibling with full or nearly full health, the other in critical at one or two squares, and the bandage goes out without a dilemma. The only two ties are the two cases in which the row of the wounded was mute, because the report the giver had said sixty-seven and was old. The one that gave without hesitating had a fresh number for the sibling's health. The two that gave on a tie only had the health band that is seen from outside, and the tie was broken by a rule of the decider, not by the other's need. It is the lesson of 4.1 seen from the other side: when what is known arrives fresh, the row converts it, and the gift goes out with a margin. There was also one case in which healing and giving competed in earnest. The giver was wounded and carried a single bandage; using it on itself and leaving it for the sibling both carried real weight in the decision. It did not see the sibling: what it had was a report from a second and a half before that put it at forty-four and without bandages. It dropped the bandage, and it was not a tie: giving won clearly. It is the case that sustains "down to its last bandage". And a limit that must be said with its figure: giving well is not coordinating well. Of the eight bandages dropped, three were picked up by the sibling, and all three while it was worse off than when they were given; the other five stayed on the ground or were surplus. And none of the three came to heal: all three were used almost at once, and in all three a blow or death cut the channel before it ended. All three died with the sibling's bandage in the bag. Of the eight scenes, the only cure that was completed was that of the one that healed with its own. The cause is not in a row: the report says what one carries, never what one has left on the ground.

A temptation that was not needed. It was also tried that being far from the sibling, in calm, be a background unease, so that they would tend to walk together. It did not serve: below a certain size it changed nothing, and from there up the only thing that changed was that the agent stopped picking up the food it had two squares away in the opposite direction. And it was not needed, because the ring already brings them together: the fire that tightens from outside ends up taking both to the same place. It was left built and switched off.

Two scenes from the diary. The diary of the pack is an account of the games of the clean series written only with what the reports say — position, health, poison, bandages, aggressor —, with no invented dialogue and without guessing what the report leaves unsaid. To read the figures: the two siblings are born with one hundred points of health; below sixty the slope begins to weigh; a sword blow from a strong rival takes almost twenty, and a blow kills when the health that remains does not reach what it takes.

The two bandages. The first is at sixteen of a hundred, where a sword blow kills it; the second, at twenty-eight, two blows away; the same aggressor, with a sword, is hitting them both. When the second falls to the wounded health band, the first has two futures before it, and only two: staying still or dropping the bandage. Healing is not among them, because of the clock fault that section 6 tells of. The two futures hurt it exactly the same, and faced with a tie the decider prefers doing to not doing. It drops, and since bandages are stacked in a single slot, both come out. The second falls a second later; the first, a little after. No one picked up anything. Two bandages of fifty on the ground, one square away, between two siblings that died at eight and at sixteen.

The one that arrives. The second is at seven of a hundred, with an unused bandage in its bag — the report says so, and the first knows it. It has tried to heal: it used a bandage, a sword blow cut it halfway, and afterwards the healing clock kept it barred for more than a second with the channel already dead; when it could try again, for almost two seconds moving was worth more than healing. The first, healthy, arrives from far away with its health whole and drops its bandage next to the other; the forty points it loses in this scene — three spear blows from the same aggressor — it loses after giving it. What moves it is not that the sibling lacks bandages, which it does not: what moves it is that it lacks health, and the row of the wounded does not look at the other's bag. A quarter of a second later, the second heals — with its own: when movement entered its cooldown, healing was left as the only future that was not staying still, and it won. No one hits it during the channel, and the bandage gives it back fifty: from seven to fifty-seven. The one on the ground it never picked up; it stayed three squares away, in sight. Why the aggressor had stopped hitting eight ticks before the sibling arrived, the log of this window does not say.

What it taught. The first scene is the limit, and it is also the weakest of the eight bandage gifts of the series: a tie, with the cure out of the vote through a fault of the agent's own — its healing clock does not notice that a blow has cut it, and it bars itself from healing for almost two seconds believing it is still healing —, and with two bandages through another fault: dropping empties the slot. The two faults are declared in section 6; they have not been corrected within this work. What the scene truly teaches is that the gift comes late when the hunter is on top, and that a gesture of care can be born at the worst possible moment and be of no use. The second scene is care at a distance as far as it goes: knowing without seeing, going, and leaving. Picking up, no: the one that was worse off healed with the bandage it already carried, and the one left for it stayed on the ground. And the whole rung taught something about rows, without characters. The first provision only lit up if the giver carried two bandages; it seemed prudent, and nothing happened in forty games. The reason was not in the row but in the rest of the agent: the row that makes the empty

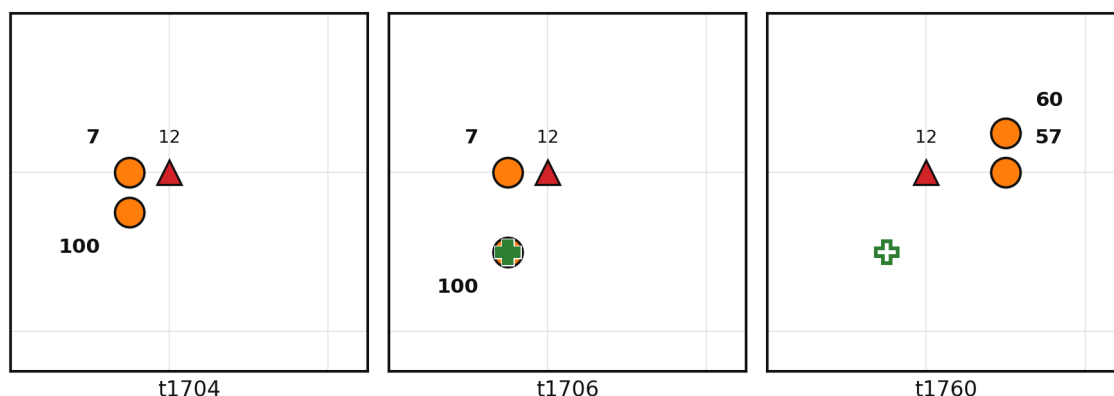


Figure 3. The one that arrives, in three instants of game ereq_b915417a, drawn from the logs. Tick 1704: the first (health one hundred) has arrived one square from the second (health seven); the aggressor, seat twelve with a spear, in red, right beside both. Tick 1706: the bandage on the ground, green cross, dropped the tick before on the first one's own square, two from the second. Tick 1760: the second has healed with the bandage it already carried, from seven to fifty-seven; the one on the ground is still where it fell, three squares away, drawn hollow because at that tick neither of the two sees it: it is the position remembered from seventeen ticks before. The first is at sixty: three spear blows from seat twelve, all after the gift. Each sibling's health sits beside its dot; the crop is the same in the three panels.

bag hurt goes quiet as soon as there is one bandage inside, so the agent almost never picked up a second. A new row does not act in a void; it acts on the situations the other rows already create or prevent, and if those situations do not occur, the new row does not occur. That is why it was changed: provision came to light up with a single bandage of one's own, when the sibling has none left. That is giving from what one also lacks, and it is what gives the rung its name.

4.3 The pack with a name

What was added. The pack of section 3: whoever is hitting my sibling presses on me too, if it is certain and if I carry a weapon that does damage. Certain meant, at first, that the sibling's fresh report said it was being hit and from what position, and that in that position there was an enemy in sight. From then on two counts are kept, and stated separately: strict initiations — hitting one that has attacked neither of the two —, which remain zero always, and defences on the sibling's behalf — the first blow against the other's certified aggressor.

What happened, first series. With that row in the table, the sibling began to defend: twelve times in seven games a sibling put itself within reach of the other's aggressor and returned the blow, and in one game the two took turns against the same hunter, each after it had struck the other. This work calls that behaviour a pack, and from it comes its title. Here the pack is two, and the word is used in its animal sense: a group whose members look after one another, warn one another and defend one another to survive. Nothing in the row counts how many they are: it looks at whoever a fresh report names as a sibling's aggressor, and with three or four it would be the same rule with more reports to read. That has not been tested; what has been tested is the smallest possible pack. The eight of the hive were already a pack in that sense, but without knowing who was who; the two here are the first pack whose members know one another. And two things appeared that the agent did not know how to do. The first is the example section 3 uses for "futures, not actions": four blows from one sibling to the other in the whole series, three of them with the spear, defending the other from an aggressor standing right behind it. The agent, on imagining its spear blow, did not include the sibling standing in

between; as soon as the image included it, the sibling's rows made that future hurt, and the spear stopped being driven through it. The second: of twenty times in which a sibling saw the other attacked, with certainty and with a weapon, in twelve the blow was physically impossible because the hunter was not within its reach, and the agent stayed still watching the fight three squares away. Just as approaching the wounded relieves a little before arriving, approaching until the sibling's aggressor is within range came to relieve a little the pressure of seeing it attacked. Neither of the two is a new row: they are two corrections of what the agent imagines.

What happened, second series. With the two corrections, in the clean series — the one that no longer had our agent alone seated among the rivals (section 3) —, the rung's headline figure: in forty games, one hundred and four defences of the sibling, zero strict initiations and zero blows between siblings. The agent defends when it can and does not wound the one it saves. What it costs it was measured on its health in the ten seconds after defending: twenty points at the median, and not one defender fallen. And then it was measured better: defending costs no more than standing beside a hunted one. That it also protects the hunted one goes in the right direction and is not shown.

What was put into the report. The aggressor's seat: the number of the one hitting me, which the one being hit knows with certainty and no one else can see. With the name in the report, the sibling no longer looks for the hunter at a position from four seconds ago but where it is now, and of the six times in which it had not defended before, in three it now defended; the other three remained without a possible blow, and rightly so. What rivals do when a sibling answers them was also looked at: no fight ended with one of the two dead, and in almost half the aggressor left after about four blows. With that picture in view, the response to blows was left as it was.

What it taught. That the care of this agent knows how to strike without striking first, and knows whom: the pack does not hunt on rumours, and does not wound the one it saves.

4.4 Memory

What was looked at first. When a hunter leaves, does it leave? The eighty games of the two series were gone through looking for returns — a seat that had already harmed one of the two and that is in sight again —, counting as aggressor only what the agent itself could know: whoever hit it, or whoever the sibling's report named. The rung's headline figure: one hundred and thirty-eight returns in eighty games, almost twice as many as there were fights. The world is small; those that hit rotate, leave and come back, with four and a half seconds at the median between going and returning, and in more than half of the returns the one coming back had hit the sibling and was now coming for the other.

The blow that was not written. The exact case was looked at — it returns, armed, it approaches — and fifteen came out. In seven, it struck first. In six, no one struck anyone. A preventive blow would have fallen, in four cases out of ten, on one that was not going to attack; and in one hundred and thirty-eight returns there was a single death of ours. It was not written. What was added was fear: that the agent be frightened before the blow on seeing a known aggressor come armed, and that what comes out of that fear — distance, wall, sibling, nothing — be decided by the body.

What was added, and what it did. A memory per game of the seats that have harmed either of the two, with the damage they did. What is remembered is a seat, not an individual: the number is valid while the game lasts, and when it ends, that seat is occupied by another and nothing remains. And a row that lights up when one of them is in sight, armed and approaching, with the size of the fear that damage would cause. It

was tested over the fifteen real situations. The row lit up when it should — thirteen of fifteen —, seven clock steps before the blow; it lit up even before one that had never touched it, because the sibling had told it the seat. And it did not change a single decision. Why, in plain words: the agent does not decide row by row. Every future it imagines carries on top of it, summed, everything that already hurts it at that moment — the health it is missing, the hunter pressed against it, the ring closing in —, and in a return those three things are usually pushing hard. A new fear the size of a blow, some eighteen points, is small beside them: it was added, but it did not change the order of the futures; the one that won before went on winning. That is why a row can light up when it should — thirteen of fifteen, seven steps before — and not move a single decision, and that was known before playing it in the field.

Remembering what it cost. The first memory kept of each aggressor only the last thing it had done: one blow, some eighteen points. On looking at the fifteen returns, those coming back were not one-blow types: between the two siblings they had taken fifty-two points at the median in the game, and one of them more than two hundred — two whole lives. The fear that comes from remembering a blow is small; the one that comes from remembering everything that seat has done in the game is real. The memory was changed to sum all the damage of that aggressor to either of the two, and it was tested again over the same fifteen situations. Fear came to move the action in eight of the fifteen, and all eight were the same thing: to stop standing still and gain distance. Without a single blow born of memory, and without a single instant in which fear made it abandon a cure or a meal. That version is the one that closes the work.

What it taught, and what it did not. It is a result over fifteen real situations, checked by making the agent decide over them; it was not played in the field, because with seven cases per series of forty no direction could have been seen. It is said so and left so. What it teaches is that this agent can remember whoever did it harm without that turning it into a hunter: the memory serves it to move away, not to strike.

The ladder, seen whole. Seen from afar, the ladder is a single story. First the agent was given a voice, and the voice was of no use, because there was nothing in it that was hurt by what the other said. Then it was given that pain and that relief — the sibling's health as its own health —, and care appeared: first up close, then at a distance, then giving what one carries before it is needed. Then it was told who was hitting the sibling, and care learned to defend without hunting. At the end it was given memory, and the fear that came out of it made it move away, not strike. On no rung was a behaviour written for it; on each one it was given something that hurt or relieved, and the behaviour came out on its own, or did not. At the end of the ladder, in the scenes of the one that goes alone — where there is no sibling in the game, neither in sight nor by radio —, the agent goes on deciding exactly the same as the one from the previous season, checked decision by decision. Everything it does anew — defending, giving, moving away from the one that returns — it does only when the other is in the scene. The whole difference between the agent alone and the agent in a pair is the other.

5. The proof by absence

The previous section told what appeared when a few rows that looked at the sibling were added to the agent. This one tells what disappears when they are taken away. It is the other half of the same test, and the one that answers the most serious objection that can be made to this work. The headline figure is a pair of numbers: zero defences of the sibling in forty games without the rows, one hundred and four with them.

5.1 The objection

To everything this work has told, one objection can be made, and it is worth stating in plain words, because it is the most serious it has. The agent does many things: it eats, it heals, it moves away, it responds to whoever hits it. How much of that is the engine, the machine that imagines the next state and chooses the one that hurts least, and how much is the table written by hand for this world? If the table does all the work, the engine is an ornament, and what is called care here would be only a well-disguised list of instructions. The objection weighs more here than anywhere else, because the ladder of section 4 added new rows, and every new row is one more occasion to say: the behaviour was written there.

There is only one way to answer with data, and it is the one the design allows: to remove the rows and look at what goes with them. If care disappears when the rows that look at the sibling are switched off, and everything else stays the same, then those rows contribute exactly care, no more and no less. If things disappear that should not — if the agent without social rows struck earlier, or struck the sibling —, then the rows would be doing more than they say. And if care does not disappear, the whole argument would be wrong. The test was done at two levels: over recorded scenes, and in the field.

5.2 Over the recorded scenes

The first thing was to make sure the rows were separable: that each one changes the value of a future only when the sibling is in it, and that in the scenes of the one that goes alone — where there is no sibling in the game, neither in sight nor by radio — the agent of the ladder decides exactly what the one that goes alone decided. That was checked at each rung, before playing it in the field, by making both versions decide over the same recorded scenes: in none of the three versions that climbed the ladder was there a single decision different from the previous one when the sibling was not in the scene. That is the weak form of the claim. It says that the new rows rewrote nothing of the one that goes alone; it does not say what they contribute when the sibling is there.

For the strong test a bare version was needed: the agent as it was at the end of the previous work, without any of the ladder's rows, seated as a pair in the same world. Before spending a single game it was checked, state by state over six scenes — the agent at full health, at half health and nearly dead, each with the sibling in front of it and without —, that this bare version decided exactly the same as the historical one, and that with the wounded sibling in front of it no social row lit up. It was also checked that, with the ladder's rows switched off, the agent of section 4 passed one by one the twelve tests of the one that goes alone. With that, what was going to be measured in the field was one thing only: the absence of the rows.

5.3 In the field: the same series, without the rows

Two copies of the bare version played forty games as a pair against the same fourteen rivals, seat by seat, that the pair of section 4 had faced in the series of the third rung, the one with the one hundred and four defences. Before playing it in the field, six things were left written that had to happen if the argument was true: that there would be no defence of the sibling; that there would be no gift next to it; that the agent's responses to whoever hits it would stay within three in ten of the series with rows, and that it would go on not striking first; that the distance between the two siblings would be greater; that there would be neither a blow nor a chosen attack against the sibling; and about the score no sign was written, because there was no reason to expect one. The following table gives the

result, measured with the same criteria in both series.

	bare pair	pair with the rows
defences of the sibling	0	104
gifts next to the sibling	1	98
own responses to aggression	711	918
responses per thousand recorded ticks	6.4	7.2
times it struck first	0	0
blows to the sibling / chosen attacks against it	0 / 0	0 / 0
median distance between the two (squares)	4.24	3.61

The left-hand column is the pair without the rows that speak of the sibling; the right-hand one, the same pair with them. Same fourteen rivals, the same number of games — not the same games —, same counting criteria. *Gift* is here an instant in which one of the two chooses to drop an object — a bandage or a ration — with the other alive and within two squares. It is a wider criterion than that of the diary of section 4, which counted only bandages; it was applied equally to both series so that the comparison would be fair, and that is why the figures do not match those given there. Five of the six things written down happened. The sixth did not: there was a gift, one only, and that gift is the subject of the next subsection. The games of one series and of the other are not paired one to one, and that limits what can be claimed about the small differences; about the large ones, it does not.

5.4 The gift that was not supposed to be there

The story this work was going to tell was clean: a body without social rows against a body with them. A single gift in forty games was enough to bring it down, and it deserves to be told in full, because it corrects something true.

It happened eighty-three seconds into a game. One of the two, with the sibling a little over one square away and at sixty health, chose to drop its rations. In the bare version there was no row from the ladder; what lit up was one of the two rows from the previous work that already looked at the partner, the one of the small unease over its harm seen up close, which that day was at half its range. That row, and the other, the grief over the partner's death, had never been switched off because they had never been counted as social. They were, in the agent of the previous season, the smallest possible version of care: a pain of one's own over another's harm, with no voice to say how much it was or whom it came from.

The honest comparison is not, therefore, without the social against with the social. It is *minimal* social against *complete* social, and that is how it is stated from here on: from one ration to ninety-eight gifts, and from zero to one hundred and four defences. The correction costs the argument nothing and sharpens it. What the ladder's rows add is not that care exists, which already existed in embryo, but its reach — the sibling's health read from the report, at a distance, instead of a neighbour's harm seen up close —, its cost — the last bandage — and its direction — the one that is hitting it.

5.5 Whether caring costs points

There remains the question this world makes louder than any other: whether caring for the sibling costs the agent points. No prediction was written, because the question was truly double. Caring could cost it dearly: the one that goes towards the sibling does not go towards a weapon or towards a weak rival, and loses time and chances. Or it could give it

an advantage: two that heal each other last longer, and lasting is what scores.

The result over forty games is this. The bare pair scored somewhat more than the complete one, a little over three points per game against a little over two and a half, and finished in the same place: the same median position in both series, and the two pairs falling together, both in the tail places, the same number of games. With forty unpaired games, that difference is below what the test can tell from chance; the detail is in the appendix.

The exact sentence the data allows, and no other, is this: caring for the sibling gave it no advantage on the scoreboard, and it is not shown that it cost it any. It matters to say it in full, without overstating it: in a world that pays for killing, what this agent does for its sibling it does not do for the scoreboard, which does not reach it. It does it because of what hurts it.

5.6 What the rows contribute, and what they do not

Three things follow from the table. The first is that defence and the gift belong to the social rows: without them there is not a single defence in forty games, and there is one gift, that of the small row. The second is that restraint does not. The bare pair strikes first exactly as many times as the complete one — none — and does not touch the sibling either, and that cannot come from rows it does not carry. It comes from what the agent alone already carried: the row that responds to whoever hits it, which is relieved by hitting that one and only that one, and the fact, told in section 3, that a blow of its own relieves only the pressure of an aggressor that is already hitting it — its imagined blow lowers that aggressor's foreseen health, and no row makes anyone else's foreseen health worth striking —, so that there is no appetite in it to attack anyone, sibling or not. What the ladder's rows add is not restraint, which was already there; it is whom to defend.

The third is that the social rows do not make the agent fight less for itself. With them lit, its responses to whoever hits it rise by a little under a third in the raw count, from seven hundred and eleven to nine hundred and eighteen, and by a little over a tenth per recorded tick — the log of the series without the rows covers less, as section 6 says —, within the margin left written beforehand and with no predicted direction. The simplest reading is that the two are closer to each other — the median distance drops from four squares and a quarter to a little over three and a half — and so are more often within the reach of whoever strikes; but with forty unpaired games that is a reading, not a measure.

It is worth saying where this leaves this work with respect to what has already been done. That an agent attend to the other's situation has been achieved before by putting it into the reward — the inequity aversion of Hughes et al. (2018) is the clean case: the comparison with the other enters the number the agent maximises —, and also by choosing altruism as the objective of optimisation without external reward (Franzmeyer et al. 2022). Here it is in neither of the two places: it is in what hurts it, and it does not touch the world's scoreboard. That is why switching off a row and looking at what disappears is, in this set-up, a question with a direct answer.

With this, the objection of section 5.1 has an answer with a size, and it is worth saying how far it reaches. The rows written by hand for this world are what care depends on: without them there is no defence, one gift, and less closeness. They are not what restraint depends on, which was already in the one that goes alone — in its earlier table and in what it imagines —; what part of that belongs to the engine and what part to the old table has not been separated, and section 6 leaves it as the test that is missing. In the score and in survival no difference was seen that forty games can tell apart. And that those rows

contribute care without saying what to do is not shown by removing them: it is shown by their form, which anyone can read, and that is the limit section 6 declares. What remains when they are removed is an agent that goes on not striking first, and that does not turn towards its sibling when it hears it fall.

6. The limits

This work has the limits of what it is: forty games per series, a single world, the same fourteen rivals, and an agent whose table its author wrote by hand. They are all stated here, in four groups: what the evidence can see and what it cannot; what is written by hand and what that means; what the agent does not know or see; and two faults of the agent that played.

6.1 What forty games can see

Each series is forty games of about three minutes, and forty games see only the large effects. Zero defences against one hundred and four can be seen; half a point of difference on the scoreboard cannot. That is why there are results that are stated with their size and not as demonstrated. That defending costs no more than standing beside is, strictly, that no difference was seen in thirty-two pairs of situations; with forty games that does not show that there is none. That the defended sibling comes out better goes in the right direction, and is not shown. And memory — the fear of the one that returns — was never played in the field: it is a result over fifteen real situations, checked by making the agent decide over them, because with seven cases per series it could not have been seen any other way. What this work claims about memory is that, and no more.

Nor are the series paired. The series with the rows and the series without them had the same rivals and the same number of games, but not the same games, and the small differences between the two mean nothing. There is a single world, a single set of rivals and a single body. Section 3 showed that changing who sits changes the world; what has been measured here holds for these fourteen, and of another fourteen nothing is known.

There is, besides, a gap in the logs themselves, said in section 3: the logs of the agents that live long do not begin at the first tick, because the platform cuts them when they reach a fixed size and goes on from zero. There is a log for six of every ten ticks the pair lived in the series with the rows and for a little over five in the series without them, and what is missing is the beginning of the long games, which are the ones that went best. That changes how two kinds of number are read. The counts — one hundred and four defences, ninety-eight gifts, one hundred and thirty-eight returns — are counts over what was recorded, and in the whole games they can only be larger. The absences — never struck first, no blow between siblings — hold over what was recorded, and of the lost stretches nothing is known: in some of them the agent did hit someone, the world's counter says so, and the log that would say whether it was answering or starting is not there. That is why the comparison of responses between the series with the rows and the series without them is made per recorded tick, and gives a little over a tenth, in the same direction as the raw count. Whether the whole log remains on the server has not been checked, because checking it means downloading again, and this work downloads nothing after closing its series.

It is also worth saying what kind of evidence this is. The first work of the series measured the agent from inside — distances, forces —; this one measures it by what it does in the field, read from the logs. They are two different kinds of evidence and they are not mixed. And what was measured here is cooperation at a cost to oneself, and only that: defending

cost health in the ten seconds that followed — twenty points at the median, though standing beside without defending cost much the same —, and giving cost the last bandage. That is what this work calls care; nothing more generous was measured.

Lastly, non-violence was possible only rarely, and it is worth saying clearly what that means. The agent does not kill by decision or by rule: there is nothing in it that killing serves, so the question is not whether it managed to restrain itself, but how far one that never attacks gets in a world where kills score points. The answer, with figures. The agent alone, before there were two, was the last one standing five times out of two hundred and forty in our own series, two of them without killing anyone, and four out of seventy-two in the public league, one without killing. As a pair, the two siblings fall into the tail places in eleven out of every forty games. Surviving to the end without killing happens, and it happens rarely: the normal thing, for an agent that does not attack, is to die before the game ends. That is what not killing costs in this world. The work does not say that non-violence wins; it says that one can live that way, and how much it costs.

6.2 What is written by hand

The table was designed by the author, and this is said plainly. Its sixteen rows and its numbers came from two hands, and each number carries the mark of which: design decisions taken before playing, and adjustments made during the work by looking at recorded scenes of real games — trying a value, seeing what the agent did, correcting. That is, put crudely, a calibration with a human as optimiser and a very small sample. There was no automatic training of any kind, but a human eye adjusting over scenes; the field series were played afterwards, and that eye had not seen them.

It is worth adding that writing that table by hand is not a shortcut of this work, but the ordinary practice of the field for decades: the cognitive structure of emotions of Ortony, Clore and Collins (1988), which is the most cited reference in the area, is also a table written by hand. The difference from the usual appraisal architectures lies elsewhere. A model like EMA (Marsella and Gratch 2009) derives the appraisal from a causal interpretation of the situation: what has happened, who caused it, whether it could be avoided. This table derives nothing: it declares. And the cut that defines this work is best seen stated against Frijda (1986), for whom an emotion already includes the tendency to act in a particular way. Here the cut is made just before that: what hurts and what relieves is declared, and what is done with it is written nowhere. That frontier is the contribution and also the limit: if someone found in the table a row that, instead of saying how much something hurts, said what to do, everything this work claims about behaviour would fall apart. That is why section 5 removes rows and looks at what goes with them, and leaves the form of the rows for anyone to read.

There remains an objection of substance that this work cannot settle. If emotions have no fixed signature — if they are constructions that vary with situation, culture and person (Barrett 2017) —, a fixed table of sixteen lines is suspect by definition. The answer that can be given is that the table neither names nor categorises emotions: it declares what hurts and what relieves along bodily axes, which is where that same theory places the regulation of the body. A reader coming from the opposite tradition, that of basic emotions with a signature of their own (Ekman 1992), will read the table differently. This work does not settle between the two, and not out of caution but because it uses the word for something else. Here an emotion is neither a category with a fixed signature nor a construction that varies with culture: it is a position in a space of three axes — a combination of how far the body, the resources and the other are displaced —, and its size is the number that moves the agent. The names — fear, care, bond — are regions of that

space that the reader may call whatever they like; the table does not name them. It is a concept distinct from that of the two traditions, and it is declared as such.

"No scripted behaviour" is, for that reason, a truth of degree, and it is said with its degree every time it is said. What is written is what hurts and what relieves, how much and on which axis; what is not written is the step from feeling to doing. The full formula this work holds is this: a homeostatic appraisal inside an architecture with memory and explicit rules — the report, the aggressor's seat and the memory of the game are written mechanisms —, where what each part contributes is measured by removing it. And "without reward" means exactly without training and without using the scoreboard. The agent does have a single criterion, the distance to its equilibrium, which is a number, and it decides by approaching it; it is not claimed that it decides without a criterion, but that that criterion is not the world's scoreboard and was not learned from it.

That distinction has an objection with a name, and it is better to state it than to wait for it. Keramati and Gutkin (2014) showed that maximising a reward and defending homeostasis are mathematically equivalent objectives; if that is so, calling this "without reward" would be a play on words. The equivalence they show holds for an agent that learns a policy by maximising that number over time: there is a value function, there are temporal differences, there is credit assigned between what was done before and what came out afterwards. This agent has none of that. There is no value function, no policy that improves with experience and no credit to assign: at every instant it imagines the states it can reach and chooses the one that hurts it least. What is claimed, then, is not that there is no number — that is already conceded —, but that that number was not learned and is not aligned with the world's scoreboard.

There is a second form of the same objection. The published rebuttal of the claim that reward is enough to explain intelligence (Silver et al. 2021) proposes a vector of rewards instead of a single one (Vamplew et al. 2022), and a reader may ask how three homeostatic axes differ from that under another name. In two sentences: the axes are not maximised nor summed with learned weights, and there is nothing that grows when things go well for the agent. They define a distance to an equilibrium, and the only thing the agent does with it is descend it.

There remains an asymmetry declared in section 3 and unresolved. The row of provision lights up when the sibling lacks bandages, is relieved by giving any cure, and what comes out of the bag is the first thing there, almost always a ration. It has not been decided whether that is a scarcity of bandages or an agent that drops the first thing it finds; telling them apart requires counting the bag at every gift, and it was not done.

And there is a test missing that this work did not do and that would be the next one: that of the engine. Section 5 removed rows and measured what each contributes; no one has yet changed the appraisal itself — the geometry of value, the way of summing pains and choosing the future that hurts least — leaving the table and the decider as they are. Until that is done, what section 5 attributes to the engine is attributed by elimination, not by direct test.

It is worth saying why that hand was necessary. This agent does not learn: nothing in it adjusts with experience how much a thing hurts it or relieves it. An animal extracts that logic on its own, by living; here a human extracted it from outside, by looking at recorded scenes and correcting. What was written by hand is not the behaviour, but the translation of this world into the agent's language: which things in ZERO-SUM hurt and which relieve, how much and on which axis. That this translation is done by a person is the limit. The ideal would be for the agent itself to extract it, adjusting game by game how much each thing in this world hurts it and how much it relieves it; today that is far off,

and it is the first piece this work would put in place next.

6.3 What the agent does not know or see

Three things about the agent's body limit what its results mean.

The first: the agent does not imagine the effect of a blow on itself. What its own blow takes, it knows, and section 3 says what it does with it; but it does not know that at sixteen health a strong rival's sword blow kills it and an arrow does not, nor that at twelve either of the two kills it. Its fear near death comes from the slope of its own health and from having someone on top of it hitting it, and it is the same with a sword in front of it as with a knife. The damage of each weapon is fixed and known, so it is an open door: it could be given. It is left as a door and not as a result, because nothing this work claims rests on the agent knowing it: the defences, the gifts and the withdrawals happened without that information. Giving it to it would change what it does near death, and that would be another result.

The second: memory is of a position, not of an individual. What the agent remembers of whoever did it harm is a seat number, valid while the game lasts; when the game ends, that seat is occupied by anyone else and nothing remains. No reputation is possible across games, because the neighbour does not repeat; and the sibling is no exception: it is a seat number that arrives once, at the start. Everything this work calls recognising, remembering and defending happens within three minutes and is erased with them. It is worth saying what kind of recognition that is, because it has a name. The agent does not recognise an individual: it recognises "the seat that did us harm in this game", which is a temporary label on a category — aggressor — and is erased when the game ends. Many species do something similar: they respond to "a predator" or to "a neighbour that behaved badly recently" without keeping a memory of individuals across a lifetime (Tibbetts and Dale 2007). It is a real form of recognition, with its properties and its ceilings, and this agent is in that box and not in the one next to it. Within the game there are also two gaps, stated in section 3: poison arrives without an author, and the camouflaged one disappears from the list of the visible; in those two cases there is no seat to fear.

The third: giving well is not coordinating well. Of the eight bandages a sibling dropped next to the other, three were picked up, all three by the one that was worse off; the other five stayed on the ground or were surplus, and none of the three healed anyone: a two-second channel does not hold under fire, and under fire is where the sibling is when it needs it. The report says what one carries and how much health one has left; it does not say what one has left on the ground, and the one that could pick it up does not know it is there unless it sees it. The gift relieves the giver whether or not the other needs it; in people that has a name — impure altruism, or "warm glow" (Andreoni 1990) — and it is one of the known reasons for over-giving. The care of this agent reaches as far as dropping, and sometimes as far as the other's hand; as far as its life, in these eight, not once. That the other makes use of it is another matter, and it is not resolved.

6.4 Two faults of the agent that played

The agent that played the series of this work carried two faults that were discovered on reading a scene — that of the two bandages, in section 4 — and that are detailed here. The first is in the healing clock: when a blow interrupts a bandage, the agent does not notice that the cure has been cut, and for almost two seconds it cannot try again, even as its health keeps dropping. The second is in the gesture of dropping: the agent values dropping one bandage, but what it drops is everything it carries in that slot of the bag, and if it carried two, both come out. The two faults affect the same thing, cures and gifts

under attack, and push in the same direction: less healing of oneself, and gifts larger than the agent valued. Neither of the two has been corrected within this work: doing so would oblige replaying the series. All the numbers given here are of the agent with the faults inside, and they are left so.

These are the limits. None has been taken to the appendix so that it is not seen, and none, stated in full, leaves without support the sentence that opens the work: in what the logs keep of forty games against these fourteen rivals, this agent never struck first, defended the sibling one hundred and four times, gave it what it carried, and none of that gave it an advantage on the scoreboard, as far as forty games can see. What the sentence claims fits within what has been measured; what does not fit has been said here.

7. What this means

7.1 What this says, and what it does not

The question at the start was whether a machine can move through a violent world without being violent: live, care for another, and sometimes get far. The answer has a size, and the size is part of the answer. An agent that never strikes first was the last one standing five times out of two hundred and forty games in our own series, and four out of seventy-two in the public league. One can live that way. One can live little.

It is worth saying carefully what kind of thing that is, because the word that first comes to mind is the wrong one. This is not a virtue. There is no temptation to fight against: there is nothing in the agent that killing serves, no pressure that a blow to someone else relieves, no future in which a stranger's death is better than its life. The agent does not restrain itself, because it has no need to. What the result shows is smaller than a virtue, and more useful: that the violence of a world and the appetite of the one that inhabits it are separable things. A world where kills score points does not oblige anyone to want to kill in order to last in it. That is all that has been shown here, and it is not little: in a world like that, the natural thing is to expect the opposite.

Whether that says something about human societies — how much of the violence in a society comes from what its members want and how much from what gives an advantage in it — is a question that accompanies this work and that this work does not answer. What it does let one see is why in us it is so hard to look at and here so easy. In a society the two things can also be read, but they are not in a single place: what hurts is written in each one, and also in the world, which would teach what hurts even if there were no one; what gives an advantage is in the laws, in the prices, in what some tell others. They are three books — oneself, the world, the others — and reading all three at once is hard. In this agent the three are reduced to the minimum and kept apart: sixteen lines for what hurts, the rules of a league for what scores points, and a report by radio for what happens to the other. That is why here they can be looked at separately, and in us they cannot.

7.2 Three echoes

What follows are three parallels that help to read what happened. They are not proof of anything: none of them rests on a measurement of this work, and no measurement of this work rests on them. They are stated because a reader coming from psychology or from ethology will hear them, and it is better to name them than to leave the reader to place them alone.

Knowing without feeling. The patients with damage to the ventromedial cortex

described by Bechara and colleagues know perfectly well which is the good option — they explain it, reason it, defend it — and cannot manage to decide (Bechara et al. 1994); in the card task, moreover, healthy subjects begin to avoid the bad decks before they can say why (Bechara et al. 1997), and from there came the somatic marker hypothesis (Damasio 1996). It is worth saying in the same sentence that that hypothesis is disputed: it has been objected that the evidence does not force its reading and that "deciding before knowing" can be explained without it (Maia and McClelland 2004; Dunn, Dalgleish and Lawrence 2006). The parallel that matters here does not depend on who is right in that dispute, because it is simpler: the information arrives intact at a system that no longer converts it into anything. The first rung of this work is that case, made on purpose and without knowing it: the agent with the report switched on, hearing its sibling's exact health, and a table that had nowhere to put it. Of eight decisions, zero changes. The lesson of the rung is that of the patient: knowing changes nothing if nothing converts what is known into pain or into relief.

The precision of help. That an animal helps another without prize and without training is documented: the rats of Ben-Ami Bartal, Decety and Mason (2011) open the door of a trapped companion and afterwards share the food. What the eight bandages of this work add to that picture is not the help, but its aim. In chimpanzees, help hits the mark — choosing the tool the other needs — only when the helper can see what situation the other is in; without seeing it, it helps unspecifically (Yamamoto, Humle and Tanaka 2012). And in people, knowing exactly what is happening to the other is not enough for help to come out: wanting to help is needed as well (Winczewski, Bowen and Collins 2016), on top of what Ickes (1993) called empathic accuracy. Helping by hearsay is helping worse. The eight bandages of section 4 can be read that way. Of the sibling's health, two different things reach the agent: an exact number, which travels by voice and can grow old, and a health band, coarse, the one seen from outside as long as the other is in sight. In six of the eight gifts the number was recent and said that the sibling had very little health left; there the future in which it dropped the bandage beat the others by a clear difference. The other two are exactly the two cases in which the number it held was old — it said sixty-seven, it did not say danger —, the row that responds to the wounded sibling stayed quiet, and dropping tied with staying still: the tie was broken by a rule that, faced with two equal futures, prefers doing to not doing, and not by the other's need. Put short: the one that gave with a margin had a number; the one that gave on a tie had only seen a health band.

The fear that moves away. In many animals, the first response to a known danger is not fighting but avoidance (Preston and de Waal 2002); and the intensity of that response is calibrated not by the last scare, but by the risk accumulated over time — it is the predation risk allocation hypothesis (Lima and Bednekoff 1999). Between the last two rungs the same thing happened. With the memory of the last blow, fear lit up in time and did not move a single decision; with the full count of each one's damage, it moved eight of fifteen, and all eight were the same thing: to stop standing still and gain distance. None went towards the blow.

7.3 Instrument and expression

Section 1 left posed a distinction between two ways in which a signal can come out of an agent, and here it is closed. One is that of an agent trained to maximise a scoreboard: if something resembling communication appears in it, that signal is an instrument of the scoreboard, and it exists as long as it scores points. The other is that of this agent. Its report cannot be that, because the world's scoreboard does not reach it by any path: it comes out because there is a body with a state and a thread that transmits it, and if

warning were of no use to it at all it would go on warning just the same.

And it is worth saying what that distinction does not say, because it is easy to hear more in it. It does not say what is inside one agent or another; it does not measure them, rank them or judge them. It says where behaviour comes from in each architecture — from a prize, or from a state —, and that is a property of the design that is checked by reading the code, not an opinion about anyone's interior. This work describes how its agent is built. Of the other agents it also describes how they are built, as far as is known, and nothing more: it does not say what is inside any of them — neither its own nor the others.

7.4 Where it would go next

Four doors, named and not opened.

The first is announced at the end of section 6: that the logic of what hurts and what relieves in a world be extracted by the agent itself, by living in it, instead of by a person watching from outside.

The second is the test that section 5 is missing: to change the appraisal itself, the way of summing what hurts and choosing the future that hurts least, leaving the table and the decision as they are, to see what remains of care when what changes is the engine.

The third is a world with a single life, where not killing truly costs and not only a place in a table. Here, dying early is dropping a few places; there it would be the end.

The fourth is the largest and the one that depends least on this work. Almost all the worlds where agents can live today are made for language agents, for reinforcement learning or for programmed behaviour: they are asked for text, given reward, or told what they have to do. A homeostatic creature lives in them on borrowed ground, and this series has spent three works building bridges so that it fits. A purpose-built world can always be made, but a world of one's own is not a shared world: no one can compare against you, and you cannot compete against anyone. The door this work would leave open is an ecosystem of social worlds — worlds with hunger, with harm, with others, with death — where creatures made in another way can practise and be measured alongside the rest.

7.5 The close

What moves this work, said for once in the first person, is a question of alignment: whether machines will end up seeing humans as rivals or as a group. The usual way of facing it is to write the values on the outside: rules that forbid, preferences that are trained, lists of what must not be done. This work looks elsewhere, and it is worth saying that this is a wager of its author and not a result: that a creature's values are not above its behaviour but beneath it, in what its body counts as harm and as relief. The wager is not original: it is that of morality grounded in the agent's physiology (Balkenius et al. 2016), and in its recent formulation, that of values understood as an affective orientation — approaching what relieves, moving away from what hurts — and not as a preference someone specifies (Sennesh and Ramstead 2026). What this work adds to that wager is not one more argument, but some games played.

The three axes of this agent are that wager made small. Looking after one's own body, looking after what is needed to stay alive, and looking after the others and the place one occupies among them: on those three things everything it does is decided. It is not claimed here that that is all we call values in a person. Something smaller and checkable is claimed: that with those three things, and without it being written anywhere what to do with the others, an agent that lives where kills score points cares.

And the world does the other half. The same three axes in a world where no one can do harm would have produced nothing that deserved to be called care: it took something hurting, and being able to hurt another. If the wager is good, a value does not live only inside the one that holds it: it lives between a body that feels and a world that gives occasions for that to matter. From there comes what this work teaches, which is minimal: an agent in which there is nothing that killing serves does not need to be forbidden to kill. It is a door, not an answer.

Supplementary material

GIF 1. The pair in a whole game. One game of the series with the rows, from the first tick to the death of the second sibling: the two siblings in orange, the ring closing in blue, and the other fourteen seats as grey dots only while one of the two can see them — a hollow dot is a position remembered for at most two seconds after it left sight. Drawn from the two agents' own logs, so the map is theirs and not the world's: no position is invented, and the dead disappear at the tick the world records. Game `ereq_292ee72d`, ticks 1 to 3408; the siblings fall at 3388 and 3408, in places eight and nine of sixteen. Sped up: fifteen seconds for two minutes and twenty of play.

GIF 2. The one that arrives. Ten seconds at real speed, the scene of section 4: the two siblings in orange with their exact health beside them, the aggressor — seat twelve, with a spear — in red, and the bandages on the ground as green crosses. At tick 1705 the healthy one drops its bandage next to the wounded one; at 1711 the wounded one, at seven of a hundred, heals with the bandage it was already carrying, and at 1760 it is at fifty-seven. The bandage on the ground no one picks up. Game `ereq_b915417a`, ticks 1560 to 1800.

The two GIFs accompany this document as supplementary material.

Data and code availability

The logs, records and scripts the appendix cites, the Spanish version of this text and the markdown source of this document are in the project repository (<https://github.com/Manelenrico/g-emv/tree/main/paper3>).

Acknowledgments and assistance statement

This work was developed with the assistance of Anthropic language models (Claude), used as a tool on three fronts: the agent's mathematics and programming, the execution and verification of the experiments, and the writing of this text. The rule of the appendix was applied to that collaboration as well: every cited number comes from a record and was verified against its source, and no claim rests on the model's memory. The design of G-EMV, the decisions of this work and the responsibility for everything written belong to the author.

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Appendix — Where each number comes from

Every claim with a figure in the body, matched with its record in the quarry (paper3_cantera/actas/), the game and the script that produces it. The rule is simple: no row without a record or file that names it, and no figure in the body without a row here.

Sections 1 and 2

The machina_1 figures come from the previous work, and it is worth saying how they are kept, because they do not follow the form of the others. Rungs two and three closed with a record; the first did not: it closed with a specification sealed before it was run and a package of results, and it is that package — frozen, with its md5 hash — that contains these figures. The closing documents of the rung are the_pack_peldano1_spec.md (md5 0bd20d8fa9641397812e02417778e0d5, sealed on 28 August 2026), the_pack_peldano1_paquete.md (md5 a16210921c63a36b1f62cfadb975a8b3) and the_pack_camino.md (md5 34e0ecf51e4552a987e8ab82c1bf9f30), where the closing of rung one on 29 August 2026 is recorded. There is no separate repository for the previous work: these files live in the root of gemv-coworld, and the first three are already in the quarry.

claim in the body	exact figure	record / source	script
eight bodies with the same geometry in machina_1	8 agents; N = 3 runs × 10,000 ticks	the_pack_camino.md (P1); paper two (DOI 10.5281/zenodo.21994358)	—
the census only looked: the life of the village with the census and without it was the same	observe-only, bit for bit	the_pack_camino.md (P1, "observe-only bit-exacto probado")	—
eight temperaments fixed before the census; the social weight runs from more to less	table of profiles per seat: 1.90 / 1.70 / 1.60 / 1.40 / 1.40 / 1.25 / 1.15 / 1.15, in a section declared NO-LEY (not a law); seal of 17 August 2026	paper3_cantera/the_pack/temperamentos_spec.md, md5 616a19719c24c10ba92fe2602db768d7; hechos/forense_censo_temperamentos.md	pregonero/run_bench_temper.sh
they were all born two or three squares from the centre	field of birth distance to the centre: 2 or 3 in all eight	the_pack_censos_crudos.md, md5 864a0bed356a5f57e4587551ca108cee; hechos/forense_censo_temperamentos.md	—
one agent mined eighty-four times and another five	ag7 = 84; ag3 = 5 (life with seed 170534803, real-mining column)	the_pack_peldano1_paquete.md, table (A), md5 a16210921c63a36b1f62cfadb975a8b3	—
temperament does not predict how much each one mines	corr(social weight, mining) − 0.25 / − 0.18 / − 0.04	hechos/forense_censo_temperamentos.md	pregonero/run_bench_temper.sh
the census saw those at the centre well and those at the periphery hardly at all	observer→observed coverage: up to 72 % in the central cluster, 1 % in the pairs at the periphery (same life)	the_pack_peldano1_paquete.md, table (B), md5 a16210921c63a36b1f62cfadb975a8b3	—

rung two: by ear, each one had news of all; the check yielded nothing	complete coverage by signed announcement; checks almost nil and hits 0	the_pack/the_pack_peldano2_acta.md	—
rung three: criterion of two squares; reliability was born different per sender; the village performed worse and then the same or better	— (no figure in the body)	the_pack/the_pack_peldano3_acta.md; the_pack/the_pack_peldano3b_acta.md	—

One caveat about the temperaments file, which is worth stating here rather than leaving it in the index: `temperamentos_spec.md` was not in the quarry when it was frozen, and it was added to it on 9 September 2026, with the addition declared in the quarry's index. The copy is literal, unedited, and its md5 hash is identical to the original's. It is a file from the previous season — the hive —, like the other rows of this block, and it touches no data from the series of this work.

Rungs two and three do have records: `the_pack_peldano2_acta.md`, `the_pack_peldano3_acta.md` and `the_pack_peldano3b_acta.md`, in the quarry's `the_pack/` folder. No figure in the body depends on them.

Section 3

claim in the body	exact figure	record / source	script
public league: 72 games with a result, 7th of 11, median place 11 of 16, last one standing 4 times, 1 without killing	72; 7/11; 11/16; 4 (5.6 %); 1 (ereq_d53c24be)	hechos/hecho_liga_publica.md; runs/debut	—
own series, season one: 240 games, last one standing 5, 2 without killing	5/240 (2.1 %); 2	hechos/hecho_v19_campeona.md	—
the ceiling of the one that goes alone: what was compared, what was found and why it stopped	80 pairs of games already played, champion against the last one from the workshop, first 1,000 ticks: pressed up (≤ 1 square) 4.6 % vs 8.1 %, paired 32-38-10, $p = 0.55$; early contact is not predicted by birthplace (corr +0.114, $n = 160$); the remaining lever — a preventive fear of strangers — required redesigning a row and its effect on placings was not demonstrable at $n = 40$	52_primer_minuto (A, ceiling written); 51 (instrument)	—

the seat is the only thing that identifies a player: of each visible one, seat and state arrive (position, health band, weapon, poison, cure); no names; the viewer's names are put there by the platform; damage arrives with the seat of whoever struck, except poison (no author) and the camouflaged (off the list); the sibling arrives as a seat once at the start	fields of visible.agents (slot + 7 of state); damage_taken "P < slot > " / "poison"; player_config teammate_slot; the platform's participants indexed by position	hechos/hecho_identidad_jugadores.md	—
the body "sight 8, agility 6, speed 5, strength 1"	INT 8 / ATH 6 / SPD 5 / STR 1	presupuesto_acta.md (34), table "la actual"	—
bandage 50 in 2 s, any blow interrupts it, poison blocks it; ration 15 in 1 s, nothing interrupts it	50 hp / 48 ticks; 15 hp / ~24 ticks	hechos/hecho_racion_venda.md	—
"one decision, with its numbers": 19 futures in the first tick, moving towards the north-west wins with 4.267 and healing stays at 4.468; in the next, with movement in cooldown, two remain — still 4.962 and healing 4.480 — and healing wins by 0.482	t1710 and t1711 of ereq_b915417a	verification count of 10 Sep over the diary of that game	—

why the step wins: the breakdown by rows	the diary keeps the detail per candidate every twelve ticks (<code>policy.py</code>, <code>DETALLE_CANDIDATOS_CADA = 12</code>), and the nearest tick with detail is 1704, where healing is worth the same, 4.468, with predicted health 57. In that tick: F-DANO 1.242 (moving) against 0.431 (healing); R-CARENCIA 0.667 against 1.0; F-4-ALCANCE 0.35-0.40 against 0.5; S-7-AGRESOR 0 against 0.5; S-8-EXPOSICION 0 against 0.3. Aggressor: seat 12, with a spear, one square away; last blow received at t1697, thirteen ticks before. At 1704 the winning step is <code>move_E</code> (4.269); at 1710, <code>move_NW</code> (4.267). The axes per future are not kept	t1704 of <code>ereq_b915417a</code>	verification count of 10 Sep; <code>codigo/appraisal_zs_v37.py</code>
what the decider imagines when moving: the end of a heading as far as the lookahead reaches, not the adjacent square; what is emitted is one step	<code>move_*</code> candidates: <code>p2 = _simula_camino(pos, dir, mundo, speed, H)</code>, with <code>H =</code> the tick's lookahead (480 in this game), advancing <code>ticks // coste</code> steps in a straight line and stopping at the first solid; <code>movs = int(dist(pos, p2))</code>, hence <code>[19,19] → [46,19]</code> with <code>movs 27</code>. <code>paso_*</code> candidates: one square, <code>movs 1</code>. <code>ir_*</code> candidates: geodesic towards a destination during the same lookahead. What is emitted after choosing a <code>move_*</code> is a single step (<code>{ "do": "move", "dir": ... }</code>)	<code>codigo/decisor_zs_v37.py</code> , lines 289-303, 411-413, 742, 786-802, 885-887; verification check of 10 Sep	—
what the agent imagines of a blow of its own: reach and damage, landed on the first body in the line; nothing of what others' blows would do to it	own damage = catalogue damage $\times (5 + \text{STR})/10$ for melee, catalogue damage for projectiles; the target is replaced by the first body in the line for <code>reach > 1</code>; the struck one's foreseen health feeds S-7-AGRESOR (aggressor) and S-HERIDO (sibling); S-VINCULO lights up when own damage $\geq$ the sibling's certain health (fresh report or the top of its health band), magnitude 1.0, flat; dodging is not predicted (declared in the code); no row reads the damage of others' weapons — the only look at another's weapon is the binary "does it do damage" of F-REENCUENTRO	<code>codigo/decisor_zs_v37.py</code> lines 347-354, 847-870; <code>codigo/appraisal_zs_v37.py</code> lines 162, 167, 1100-1112, 1028, 1138-1139, 1381, 815; verification check of 12 Sep	—

the map of versions and series in 3.5	v29 (55) · v30 (56, 57) · v31 (59, 60) · v32 (61) · v33 built and switched off (62) · v34 (63, 64) · v35 (65, 66) · v36 (68) · v37 (70, 71) · v27 as a pair (72). Roster by ranking up to 64 (record 55 says so literally; record 66 confirms it by explaining how our champion was excluded); explicit roster of 10 UUIDs from 66 on, repeated to fill 14 seats, with 0 participants of ours outside seats 10 and 11. Cold check of 10 Sep, record by record: fourteen match; record 62 names v33 only as the arm tried and rejected and leaves v32 as the one in force, and that is how the table states it	records 55-72; 66_manada2 and 72_ablacion (roster verified)	—
the row of health: linear until losing 4 in 10, slope below 60	$M = u$ if $u \leq 0.4$; $M = u + 0.4((u - 0.4)/0.6)^2$ if $u > 0.4$	codigo/appraisal_zs_v37.py, CUESTA_UMBRALE_U, CUESTA_GANANCIA	—
with our agent alone in the game, the siblings responded to aggression a third as often	327 (64) vs 918 (66) responses to own aggressor	64_manada_campo, 66_manada2	manada_campo.py::honor_dos
the rate of defence rose from three in four to nine in ten without changing the agent	P2' 78 % (64) / 92 % (66) / 86 % cumulative	68_nombre	desenlaces.py::ocasiones_v2
census temperaments: social weights per seat 1.90... 1.15; temperament does not predict mining	corr(w_s, mining) -0.25 / -0.18 / -0.04	temperamentos_spec.md (seal 17 Aug); hechos/forense_censo_temperamentos.md; pregonero/run_bench_temper.sh	—
the logs do not cover whole games: a log for six of every ten ticks the pair lived; the beginning of the long games is missing	coverage 59.8 % (64), 59.0 % (66), 52.4 % (72); logs without a header 17 / 20 / 22 of 80; games affected 11 / 13 / 13 of 40; ceiling 10.78 MB (maximum over 836 logs); cause: the platform's capture of stdout, with purging (policy.py writes without a cap; the download does not cut); whether the server keeps the whole log, unchecked	paintball/cobertura_diarios_acta.md (md5 b71bde537d06e31122eb6070d82af285)	cobertura_diarios.py

figure 1 (viewer)	four screenshots of ereq_292ee72d at 0:00, 1:11, 2:06, 3:54 = ticks 1, 1704, 3024, 5616 (24 ticks/s); the pair's reports legible in the viewer and checked against the logs (1:11: P10 T1681 [30,17] hp 78; 2:06: P11 T3025 [25,21] hp 48); supports no data	paper3/figuras/ficha_figuras.md; md5 cdfcd19575a3e33fd7a7f31395780d89	screenshots by Manel; assembly tira_visor.py
figure 2 (logs)	ereq_292ee72d, t3029; ring r = 15 (24→19 at t2064, 19→15 at t2976); 9 alive; 6 seen + 1 remembered + the two	ficha_figuras.md; md5 75d0b626ba768c151a668ede26f16008	figuras_paper3.py

Section 4

claim in the body	exact figure	record / source	script
of eight decisions, the report changed zero	0 of 8 (only S-DANO-PAREJA ± 0.03)	53_parte (P4)	verifica_53.py
the margin for attacking the partner-aggressor drew closer	+ 0.405 $\rightarrow$ + 0.047 at h10; 0 initiations in 53 decisions	53_parte (P5)	verifica_53.py
two complete deliveries seen in the first series	ereq_331e254f (t353→t419→t607), ereq_2502e92c (t387→t430→t479)	55_pareja, 56_cuidado	—
the step towards the sibling won one decision in a hundred — CHECKED	ir_pareja chosen in 1 % of ticks with a decision (grows 189 / shrinks 185 ticks)	56_cuidado (B, forensic of the separation, 79 diaries)	separacion.py
from being within two squares of the sibling in danger on two occasions in a hundred to one in three — CHECKED	≤ 2 squares with one in the danger health band: 2 % (series 55, v29) $\rightarrow$ 36 % (series 57, v30)	57_pareja2 (P3)	pareja2.py
the gift comes late: gifts in 18 % of episodes with the condition	13 drops / 17 cond. eps / 18 %	57_pareja2 (P2)	pareja2.py
carrying two bandages: six instants in a thousand	614 of 106,066 ticks (0.6 %)	60_pareja3	pareja3.py
the give/use diagonal	table my_hp $\times$ its_hp (95...10 $\times$ 8...95)	61_dar (P3)	verifica_61 / radiografia

company: below 0.22 it changes nothing; above it loses the ration at 2	sweep 0.05 → 0.5	62_compania	verifica_62.py
nine in ten gifts were rations	90 rations + 8 bandages of 98 (criterion of 72: drop within ≤ 2 of the living sibling)	72_ablacion; hechos/hecho_racion_ venda.md	—
"eight bandages": bandage drops in 66	10 bandage drops in total; 8 within the criterion (≤ 2 of the living sibling); the remaining 2 outside the criterion	hechos/forense_dones_venda.md	—
six bandages by a clear margin (row of the wounded), two ties with the old report (67)	6 / 2; detail per game and tick	hechos/forense_dones_venda.md	—
healing against giving with provision lit: gave by a margin (report of 1.5 s, sibling at 44 without bandages)	1 case; game and tick	hechos/forense_dones_venda.md	—
of eight bandages, three picked up (all three while worse off); five on the ground or surplus; none of the three came to heal	3 / 5; 0 of 3 cures completed	hechos/forense_dones_venda.md; count by events of 10 Sep in hechos/forense_ dones_venda_addendum.md	—
THE TWO BANDAGES: two healing attempts cut in the previous 10 s; drop/still tie; stack of 2 in one slot; deaths at 8 and 16	exact ticks and values	hechos/forense_dos_vendas.md; ereq_9869b665	—
mechanical faults declared: healing clock (bars up to 41 ticks after an interruption) and dropping empties the stack	41 ticks; not corrected within this work	hechos/forense_dos_vendas.md	—
first series of the pack: 12 defences in 7 games; the two taking turns	12; ereq_feafe3ac t534/t573/t583	64_manada_campo	manada_ campo.py

four blows between siblings, three with the spear	4; 3 friendly fire in defence (spear, range 2)	64_manada_campo, 65_alcance	manada_campo.py::golpes_entre
of twenty occasions, in twelve the blow was impossible	20 windows; 12 without reach; $5/8 = 62\%$	64_manada_campo (autopsia P2), 65_alcance (P2')	autopsia_manada.py
104 defences, 0 strict initiations, 0 blows between siblings (40 games, over what was recorded; see the coverage row in section 3)	918 / 104 / 0 / 0	66_manada2; VERIFICACION_66.md	manada_campo.py::honor_dos
what defending costs in health: – 20 at the median, 0 defenders fallen	– 20 hp; 0 fallen in 240 ticks	64, 66	manada_campo.py::precio
defending costs no more than standing beside	– 14 (defends) vs – 17 (accompanies), 15-12-5, $p = 0.70$; 32 clean pairs	67_proteccion	proteccion_pareada.py
protecting the hunted one: right direction, not shown	Δhp hunted 0 vs – 14, 15-8 and 13-7, $p = 0.21 / 0.26$	67_proteccion	proteccion_pareada.py
the six times without defence: no possible blow; three recover it with the name	0 candidate in 39 ticks; 3 of 6 with identity	67_proteccion, 68_nombre (P1)	verifica_68.py
no fight ended with one of the two dead; in almost half the aggressor left after ~4 blows	74 windows: 0 % kill us; 46 % leaves (4 blows at the median); 3 % harassment	68_nombre (E)	desenlaces.py
138 returns in 80 games, almost twice as many as fights	138; 186 % of 74; median 1/ep; 51/80 eps	69_reencuentros	reencuentros.py
four and a half seconds between going and returning	median gap 110 ticks (24 ticks/s)	69_reencuentros	reencuentros.py
more than half: hit the sibling and comes for the other	53 %	69_reencuentros (P4)	reencuentros.py
exact case: 15; it first 7; no one 6; one death in 138	15 / 7 / 6; 1 death	69_reencuentros	reencuentros.py; runs/casos_exactos.json

memory keeps a seat, not an individual: valid while the game lasts	slot stable by token during the game; no persistent identifier across games	hechos/hecho_identidad_jugadores.md	—
the row lights up 13 of 15, seven steps before, and changes no decision	13/15; 7 ticks; 0 changes; magnitude 0.18 (median 0.17)	70_memoria	verifica_70.py
accumulated damage of those that return: 52 at the median, more than 200	median 52, max 211, min 8; last blow 18	71_memoria2; runs/casos_exactos.json	—
the action changes in 8 of 15, and all eight are to stop standing still and gain distance; 0 blows; 0 cures or meals abandoned	all 8: starting action <code>noop</code> (staying still) → new action <code>move_W</code> (one step); the row that diverges is F-REENCUENTRO; 0 blows out of 90 actions reviewed; 0 ticks abandoning loot or cure against 28 ticks gaining distance. The label of the sealed prediction was "distancia/pared" (distance/wall): "wall" (interposing an obstacle) is one of the outcomes anticipated in record 70 ("lo que salga del miedo —distancia, pared, hermana, nada— lo decide el cuerpo") and does NOT appear in the log; all eight are a step. "Gaining distance" is what was measured (P3); "moving away from the one that was coming" is the record's reading, not a tick-by-tick check of the geometry	71_memoria2 (P2, P3, P6); 70_memoria (prior list)	verifica_71.py
at what moment of the scene each of the eight diverges	the action diverges in the second of the six snapshots in six of the eight cases, in the third in one and in the sixth in another; in none in the first. READING NOTE: the index the script prints is zero-based over the six snapshots of ticks 300-305, so that its "1" is the second snapshot (tick 301). The TEXT of record 71 says "en el primer tic en siete de los ocho casos" (in the first tick in seven of the eight cases) and is wrong in both directions: if "the first tick" is the first snapshot, it happens zero times; if the index's 1 was read as "the first", it would be six. The table rules, corrected by the count	71_memoria2 (P6); count of 9 September 2026, two runs with an identical table (the bench declares determinism 3/3)	verifica_71.py

in the scenes of the one that goes alone, with no sibling in the game, it decides the same as the previous season	v35 29/29, v36 30/30, v37 32/32 byte for byte	65, 68, 71	verifica_65/68/71.py
figure 3 (logs)	ereq_b915417a, t1704 / 1706 / 1760; crop x[16,25] y[15,24]; bandage dropped at t1705 at [19,21]; at t1760 the bandage is in sight of neither (11 stopped seeing it at t1743) → hollow	ficha_figuras.md; md5 9d7c2ab9ee80d3e82b73f138eb70e7af	figuras_paper3.py
the one that arrives: the first arrives with its health whole and loses forty after the gift	seat 10: hp 100 at t1680, 1694, 1704, 1705, 1711; blows from P12 at t1717, t1737, t1757 (− 13.2 each) → 86, 73, 60. The earlier sentence "loses fourteen on the way" was wrong: the − 14 is the median of record 67, another magnitude	log of seat 10; verification check of 11 Sep	—
damage per weapon: sword 18 and spear 12 base × strength; bow 14 and dart 4 fixed; no loss with distance; poison 2/s; this body 11 / 7 / 14; strong rival up to 20 (19.8 seen)	—	hechos/hecho_dano_armas.md	—
reaches: sword 1, spear 2, bow 8, blowgun 6	1 / 2 / 8 / 6. The blowgun's reach comes from the world's catalogue, not from our own measurement: no squares of a dart's flight have been counted	hechos/hecho_dano_armas.md	—

Section 5

claim in the body	exact figure	record / source	script
the three versions of the ladder decide the same as the previous one with no sibling in the scene	v35 29/29, v36 30/30, v37 32/32 byte for byte	65, 68, 71	verifica_65/68/71.py
the bare version decides the same as the historical one, state by state over six scenes	v27-pareja vs historical v27: 6 scenes (hp 100/60/25 × with/without sibling), 0 differences, 0 social rows live with the wounded sibling in front	72_ablacion (prior gate)	—
with the ladder's rows switched off, the agent passes the twelve tests of the one that goes alone	benches 13-28, switches OFF, 12/12	72_ablacion	—
same fourteen rivals, seat by seat	roster identical to 66	72_ablacion; REPRO.md	—

six things written before running	P1 0 defences; P2 0 gifts; P3 responses $\pm 30\%$ and 0 initiations; P4 greater distance; P5 score without a sign; P6 0 blows / 0 attacks on the sibling	72_ablacion	—
table: defences 0 / 104	0 (72) vs 104 (66)	72_ablacion; 66_manada2	manada_campo.py::honor_dos
table: gifts 1 / 98 (criterion: drop a bandage or ration with the living sibling within ≤ 2)	1 in 1 ep (72) vs 98 in 13 eps (66)	72_ablacion	—
table: own responses 711 / 918; "rise by a little under a third in the raw count, and by a little over a tenth per recorded tick"	+ 29 % with rows (– 22.5 % without) in the raw count; per recorded live tick 0.007205 (66) vs 0.006445 (72), + 12 %; within the sealed $\pm 30\%$; no predicted direction	72_ablacion; paintball/ cobertura_diarios_acta.md	manada_campo.py::honor_dos; cobertura_diarios.py
table: struck first 0 / 0; blows to the sibling 0/0; chosen attacks 0/0	0 / 0 / 0 / 0	72_ablacion	manada_campo.py::golpes_entre
table: median distance 4.24 / 3.61 squares	4.24 (72) vs 3.61 (66)	72_ablacion	—
the single gift: at 83 s, sibling a little over one square away and at 60 health, drop rations, row of the partner's harm at half	ereq_9c3c18ad, tick 2002 (83.4 s), seat 11, soltar_rations, sibling hp 60 at 1.4 squares, S-DANO-PAREJA = 0.50; S-MUERTE-PAREJA also live in v27	72_ablacion; PROMPT_12 (season one, record 12)	replay fd29a144
whether caring costs points: "a little over three against a little over two and a half", same median, fall together the same number of games	mean score 3.12 (72) vs 2.65 (66); median placing 12 vs 12; kills 10 vs 9; better of the two in top-4: 8 (20 %) vs 5 (12 %); both ≥ 13 : 11 vs 11; one standing among the last 8: 14 vs 12	72_ablacion (P5)	—
"below what the test can tell apart"	house criterion R12: 40 unpaired games	72_ablacion; varas	—

Section 6

claim in the body	exact figure	record / source	script
half a point of difference on the scoreboard cannot be seen	3.12 vs 2.65 ($\Delta 0.47$), 40 unpaired games	72_ablacion (P5)	—
defending costs no more than standing beside: no difference seen in 32 pairs	– 14 vs – 17, $p = 0.70$	67_proteccion	proteccion_pareada.py
the defended sibling comes out better: right direction, not shown	Δhp 0 vs – 14, $p = 0.21 / 0.26$	67_proteccion	proteccion_pareada.py
memory: fifteen real situations, seven cases per series	15 exact cases over 80 games (64 + 66)	69, 70, 71	reencuentros.py; verifica_70/71.py
defending cost twenty of health at the median	– 20 hp; 0 fallen	64, 66	manada_campo.py::precio

with the rows lit it responded MORE to whoever hit it	918 vs 711 (+ 29 % with rows); direction not predicted; within the sealed ± 30 %	72_ablacion	manada_campo.py::honor_dos
last one standing 5 of 240 (2 without killing); public league 4 of 72 (1 without killing)	5/240; 2; 4/72; 1	hechos/hecho_v19_campeona.md; hechos/hecho_liga_publica.md	—
the two fall into the tail places in 11 of 40	both ≥ 13 : 11 (66) and 11 (72)	66, 72	—
two hands: [Manel] / [impl] per constant	table of signs v37 with tags	codigo/appraisal_zs_v37.py; tabla_signos_zero_sum_en_curso	—
engine ablation not done	—	future work	—
at 16 health a strong sword blow (19.8) kills it and an arrow (14) does not; at 12, both	19.8 / 14	hechos/hecho_dano_armas.md	—
memory of a position: seat valid only within the game; poison without an author; the camouflaged off the list	slot; "poison"; camouflage	hechos/hecho_identidad_jugadores.md	—
three of eight bandages picked up, all three by the one that was worse off; none came to heal	3 of 8 picked up (by pickup event); 0 of 3 cures completed — channel cut by a blow in ereq_3f9b199e (P14, t1172, 15 $\rightarrow$ 2) and in ereq_a62f7c2d (P2, t2477, 23 $\rightarrow$ 10), and death inside the channel in ereq_9fab619d (t2436, 24 of 48 ticks); all three died with the sibling's bandage in the bag. The only completed cure of the eight scenes is that of ereq_b915417a, and it was with its own bandage, not with the gift	hechos/forense_dones_venta.md; count by events of 10 Sep in hechos/forense_dones_venta_addendum.md	—
healing clock: almost two seconds unable to retry	bar up to 41 ticks (1.7 s)	hechos/forense_dos_vendas.md	—
dropping empties the slot: both bandages come out	stack of 2 in one slot	hechos/forense_dos_vendas.md	—

Section 7

It brings no figures of its own: those that reappear refer to the row where they already are, and they are listed here so that it can be seen that none is loose.

claim in the body	where its row is
last one standing five of two hundred and forty and four of seventy-two	section 6, row "last one standing 5 of 240 ... 4 of 72"
six gifts with a clear margin and two ties with the old number (it said 67)	section 4, rows "six bandages by a clear margin..." and "eight bandages'..."
eight of fifteen, and all eight to stop standing still and gain distance	section 4, row "the action changes in 8 of 15..."

sixteen lines that can be read	section 3, row "the row of health..."; table of signs v37
of eight decisions, zero changes (the report that changed nothing)	section 4, row "of eight decisions, the report changed zero"

What remains to be closed

Two things. The first supports no claim in the body: the caption of a figure for section 5, if it is added; the replay of the contrast (eda0b7d3) has not been viewed, and until it is viewed it is not described. The second does touch the body, and is said in sections 3 and 6: whether the platform's server keeps the whole log of the cut games. Checking it means requesting the logs of an episode again, and it has not been done; if it were done and they were there, the counts could be completed and the absences checked over the whole games.

Note on "distance or wall"

The label appeared twice in the body — in 4.4 and in section 7 — and does not describe what is in the log. It came from a list written before looking, in the record of the previous rung, where it was left stated that "lo que salga del miedo —distancia, pared, hermana, nada— lo decide el cuerpo" (whatever comes out of fear — distance, wall, sibling, nothing — the body decides; the record, written before the vocabulary was fixed, says "hermana"). What came out were eight steps: in all eight situations the agent went from standing still to moving, gaining distance, and in none did it interpose an obstacle. Both appearances are corrected.